ENGINEERING MANY-BODY QUANTUM DYNAMICS IN RADICAL PAIR MAGNETORECEPTION

by

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ABSTRACT

Radical pair systems provide a useful platform for studying quantum dynamics in complex, room-temperature molecular environments, with relevance to both magnetoreception and the broader goal of engineering controllable spin-dependent processes in biology. In this work, a fully quantum mechanical model of radical pair dynamics is developed in which the two electron spins and their surrounding nuclear environments are treated explicitly as a composite many-body system. The model incorporates Zeeman, anisotropic hyperfine, exchange, dipolar, and quadrupolar interactions, together with a Markovian open-system treatment based on the Gorini-Kossakowski-Sudarshan-Lindblad framework.

The results show a clear separation of timescales between coherent spin evolution and biologically motivated Markovian relaxation, indicating that spontaneous-emission-type dissipation is too weak to significantly influence the dynamics on microsecond timescales. In this regime, radical pair behavior is dominated by intrinsic Hamiltonian structure rather than external bath-induced relaxation. Comparison with the semiclassical Overhauser-field approximation further shows that, although coarse-grained observables such as time-averaged singlet population can appear similar, semiclassical models fail to reproduce the structured temporal and frequency-domain behavior of the full quantum system.

A central finding is that the nuclear environment acts not as a passive classical background, but as a finite, structured quantum subsystem capable of storing and returning quantum information. This leads to entanglement exchange, revivals, and multi-frequency dynamics that are absent in semiclassical descriptions. Exchange and dipolar couplings are shown to qualitatively reshape both correlation transport and singlet dynamics, while localized nuclear operations reveal distinct control roles for strongly and weakly coupled nuclei. Finally, a quantum-information-theoretic model of nuclear decay is introduced to examine how localized non-unitary processes propagate through the spin network and alter electronic observables.

Taken together, these results support a view of radical pair systems as mesoscopic quantum networks in which internal environmental structure and quantum correlations play a central dynamical role, requiring a fully quantum mechanical treatment for accurate description and future engineering applications.

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LIST OF SYMBOLS

GKSL dissipation rate . . . . . γ_k

Lindblad dissipator . . . . . $\mathcal{L}(\rho)$

Lindblad jump operator . . . . . L_k

Overhauser field . . . . . $\mathbf{B}_{\mathrm{OH}}$

Pauli matrices . . . . . $\sigma_x, \sigma_y, \sigma_z$

Zeeman angular frequency . . . . . ω_Z

Zeeman frequency . . . . . f_Z

antineutrino Hilbert space . . . . . H_ν

decay amplitudes in the two allowed spin channels . . . . . a, b

decay-induced entanglement / depolarization parameter . . . . . q

density matrix / density operator . . . . . ρ

dimensionless dissipation ratio . . . . . ϵ

dipolar Hamiltonian . . . . . H_{dip}

dipolar interaction strength . . . . . D

electron Hilbert space in decay model . . . . . H_e

electron Zeeman Hamiltonian . . . . . H_Z

electron gyromagnetic ratio . . . . . γ_e

electron spin components . . . . . S_x, S_y, S_z

electron spin operators . . . . . $\mathbf{S}_1, \mathbf{S}_2$

electronic singlet state . . . . . S

electronic triplet states . . . . . T_{+1}, T_0, T_{-1}

exchange Hamiltonian . . . . . H_{ex}

exchange interaction strength . . . . . J

external magnetic field . . . . . $\mathbf{B}$

final time-averaged singlet population over total runtime T . . . . . $\bar{P}_S(T)$

hyperfine Hamiltonian . . . . . H_{hf}

hyperfine coupling tensor . . . . . $\mathbf{A}_{jk}$

input Hilbert space for nuclear decay model . . . . . H_{in}

instantaneous singlet population . . . . . $P_S(t)$

isometric decay map V

joint density matrix of system and environment . . . . . ρ_{SE}

magnitude of external magnetic field . . . . . B_0

nuclear quadrupole Hamiltonian . . . . . H_Q

nuclear spin components . . . . . I_x, I_y, I_z

nuclear spin operator for nucleus k coupled to electron j . . . . . $\mathbf{I}_{jk}$

output Hilbert space for nuclear decay model . . . . . H_{out}

partial trace over emitted electron and antineutrino . . . . . $\mathrm{Tr}_{e\nu}$

partial trace over environment . . . . . Tr_E

principal-axis hyperfine tensor components . . . . . A_x, A_y, A_z

projector onto the singlet subspace . . . . . Π_S

proton Hilbert space . . . . . H_p

quadrupole tensor . . . . . $\mathbf{Q}_{jk}$

radical-pair separation . . . . . r

reduced density matrix of the remaining proton . . . . . ρ_p

reduced density matrix of the system . . . . . ρ_S

running time-averaged singlet population . . . . . $\bar{P}_S(t)$

system Hamiltonian . . . . . H_S

total electron spin operator . . . . . $\mathbf{S}_{\mathrm{tot}}$

total spin quantum number in decay model . . . . . J_{tot}

total system Hamiltonian . . . . . H

trace operation . . . . . Tr

unit vector connecting the two electrons . . . . . $\hat{\mathbf{r}}$

unitary time-evolution operator . . . . . $U(t)$

von Neumann entropy of subsystem A . . . . . $S(\rho_A)$

LIST OF ABBREVIATIONS

Density Functional Theory	DFT
Fast Fourier Transform	FFT
Flavin Adenine Dinucleotide	FAD
Gorini–Kossakowski–Sudarshan–Lindblad	GKSL
Kilohertz	kHz
Megahertz	MHz
Microsecond	µs
Microtesla	µT
Millitesla	mT
Nanometre	nm
Overhauser	OH
Quantum Mechanical	QM
Radical Pair	RP
Radical Pair Mechanism	RPM
Radiofrequency	RF
Root-Mean-Square	RMS
Tryptophan	Trp

STATEMENT ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE

Generative artificial intelligence tools were used in a limited and supervised capacity during the development of this thesis. Their use was restricted to support tasks such as drafting or debugging computational code, assisting with figure-generation workflows, checking wording and readability, and helping identify potential factual, citation, or formatting issues for subsequent manual verification.

The intellectual contributions of this thesis, including the formulation of the research aims, development of the physical models, interpretation of results, and presentation of the final arguments and conclusions, are entirely my own. Any material produced with AI assistance was reviewed critically, verified independently, and revised as necessary before inclusion. Responsibility for the content of this thesis rests solely with the author.

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CHAPTER 1

INTRODUCTION

Shortly after the development of quantum mechanics in the early twentieth century, questions began to emerge regarding its relevance beyond atomic and condensed matter systems, extending into the domain of living organisms. In a pioneering 1932 work, Pascual Jordan speculated that quantum mechanical principles may play a fundamental role in biological and even psychological processes [1]. This perspective was further developed in Schrödinger's influential 1944 monograph *What is Life?*, which proposed that the stability and functionality of biological systems may ultimately be rooted in quantum mechanical laws [2]. These early ideas laid the conceptual foundation for what is now known as quantum biology.

In the decades since, the field has evolved from speculative inquiry into a growing body of experimentally and theoretically supported phenomena. Evidence for quantum effects in biological systems has been identified in processes such as excitonic energy transport in photosynthetic complexes, quantum tunneling in enzymatic reactions, and spin-dependent chemical reactions in radical pair systems [3–6]. These developments establish that quantum mechanics is not merely present in biological systems, but can play a functional role under ambient, warm, and noisy conditions.

Two open questions now sit at the state-of-the-art in quantum biology, namely: how quantum mechanical are these bio-systems? and can we leverage our knowledge of quantum mechanics to engineer new, helpful bio-systems? This work deals with the latter. We aim to study quantum many-body-open systems grounded in biological contexts to gain insights into the quantum dynamics at play. The eventual goal being to apply these insights to engineer new quantum systems helpful in biological and medical fields.

This work focuses on the radical pair mechanism (RPM) as a structured many-body quantum system whose dynamics may be understood, tested, and potentially engineered. Although the RPM is best known as the leading theoretical framework for avian magnetoreception [4, 5, 7, 8], its importance here is broader than that specific biological setting. Radical pair systems provide a natural platform in which coherent spin dynamics, environmental structure, and experimentally relevant observables are intrinsically linked, making them an unusually rich setting for studying quantum control, information flow, and the role of finite environments in open-system dynamics [9–11].

The avian-magnetoreception literature provides the most mature physical and chemical foundation for this program. In particular, the flavin–tryptophan radical pair in cryptochrome offers biologically motivated energy scales, coupling strengths, and molecular geometries with which to ground the modelling [4, 12, 13]. At the same time, the central goal of this thesis is not to determine the minimal quantum

resources required for bird navigation, but to use the RPM as a realistic and well-structured model system for exploring how radical-pair dynamics can be understood, controlled, and potentially engineered for broader applications in biology and quantum technology [14–21].

From this perspective, avian magnetoreception serves primarily as an anchor for parameter selection and physical motivation rather than as the sole target of the analysis. The emphasis throughout this thesis is therefore placed on the engineerability of the RPM: when fully quantum mechanical descriptions are necessary, how internal nuclear environments reshape the dynamics, where semiclassical approximations fail, and which features of the system may be used as control channels in future artificial or biologically inspired implementations [9–11, 22, 23].

1.1 Quantum Information Perspective on Biological Systems

A natural framework for addressing these questions is provided by quantum information theory, which offers a language for quantifying coherence, correlations, and information flow within complex many-body systems [24]. From this perspective, biological processes may be viewed not only as chemical reactions, but as dynamical networks of quantum information.

This viewpoint enables the study of how quantum correlations are generated, redistributed, and potentially lost within structured environments. In particular, it allows one to move beyond semiclassical approximations and instead treat biological systems as composite quantum systems composed of interacting subsystems. This raises the possibility that internal degrees of freedom—such as nuclear spin environments—may play an active role in shaping system dynamics, rather than acting solely as sources of noise [9, 10].

To describe this system, we adopt the density matrix formalism of quantum mechanics, which provides a natural framework for treating both pure and mixed states [25]. Each spin degree of freedom is modeled as a finite-dimensional quantum system: the electron spins as qubits (spin- 1/2) and, where relevant, nuclear spins as either qubits or qutrits (spin-1 nuclei). The full radical pair system is then constructed as a tensor product of these individual subsystems, forming a composite many-body Hilbert space.

This formalism enables a unified treatment of coherent evolution, statistical mixtures, and subsystem dynamics. In particular, it allows the full system to be evolved under a well-defined Hamiltonian while simultaneously tracking reduced dynamics of subsystems through partial trace operations. This is essential for studying how information and correlations flow between the electronic and nuclear degrees of freedom, and for quantifying phenomena such as entanglement generation, decoherence, and environmental back-action [25].

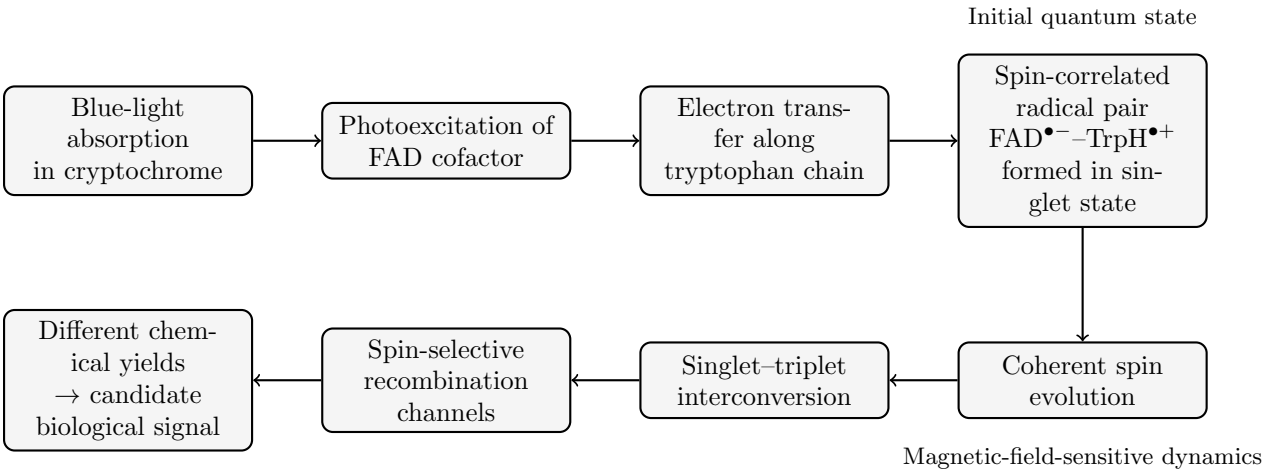


Figure 1.1 Schematic pipeline of the radical pair mechanism (RPM) as proposed for avian magnetoreception. Blue-light excitation of cryptochrome initiates electron transfer from a nearby tryptophan chain to the FAD cofactor, producing a spin-correlated radical pair. The resulting electron spins undergo coherent singlet–triplet evolution under competing spin interactions, and the spin state modulates the recombination pathway and resulting chemical yields.

1.2 Radical Pair Chemistry and Biological Motivation

A particularly compelling system in which to explore these ideas is avian magnetoreception. Many migratory bird species are able to sense the Earth's magnetic field and use it for navigation over long distances. One of the leading hypotheses for this ability is the radical pair mechanism (RPM), in which spin-dependent chemical reactions are modulated by weak magnetic fields on the order of the geomagnetic field strength ($\sim 47\,\mu$T) [4, 5, 7, 8].

The biological realization of the RPM is widely believed to occur within cryptochrome proteins. Upon absorption of blue light, the flavin adenine dinucleotide (FAD) cofactor is photoexcited and initiates an electron transfer process along a chain of nearby tryptophan residues, leading to the formation of a spin-correlated radical pair of the form $FAD^{\cdot -}$–$TrpH^{\cdot +}$ [12, 13]. The radical pair is typically initialized in a singlet state due to spin conservation during electron transfer [4].

Following their creation, the electron spins undergo coherent evolution under the combined influence of several competing interactions. Hyperfine coupling to surrounding nuclear spins induces singlet–triplet mixing, while the external geomagnetic field introduces Zeeman splitting that modulates this evolution [4, 8]. Exchange and magnetic dipolar interactions between the electrons further modify the dynamics, depending sensitively on the spatial separation and relative orientation of the radicals [26].

In addition to these dominant interactions, further contributions arise from nuclear Zeeman and quadrupolar terms. While typically smaller in magnitude, these interactions can introduce additional

structure into the nuclear dynamics and influence long-time behavior [4]. Their inclusion becomes relevant in the spirit of developing a fully quantum mechanical description of the system.

The time-dependent spin state determines the probability of recombination through singlet or triplet channels, leading to different chemical products. Experimental evidence supporting this mechanism includes the disruption of avian magnetoreception by weak radiofrequency fields, which is consistent with coherent spin dynamics at characteristic Zeeman and hyperfine frequencies [27, 28].

The relevant energy scales place the system in a regime where quantum effects can play a significant role. The geomagnetic field corresponds to a Zeeman frequency of $\sim 1.4\,\mathrm{MHz}$, comparable to weak and intermediate hyperfine couplings. At the same time, density functional theory calculations indicate that the FAD–tryptophan radical pair is embedded within a structured nuclear environment consisting of multiple anisotropically coupled spins spanning several orders of magnitude in coupling strength [12].

1.3 Theoretical RPM Modelling

Despite the complexity of the underlying physics, many theoretical treatments of the radical pair mechanism adopt simplified descriptions. These simplifications are made to capture certain phenomenological behavior of the system whilst keeping the exponential growth of the Hilbert space with system size at bay [7, 13, 29]. The most prominent simplification is the use of very few nuclei in the hyperfine environment at idealized hyperfine strengths [4]. Another common approach is the Overhauser field approximation, in which the discrete nuclear spin environment is replaced by an effective classical magnetic field acting on the electron spins [7, 30]. Additionally, environmental effects are often neglected, with spin-selective recombination typically assumed to be the dominant mechanism governing coherence loss [4, 31].

While these approximations can reproduce certain experimental features, they inherently discard key quantum mechanical aspects of the system. In particular, they neglect electron–nuclear entanglement, discrete many-body dynamics, and the possibility of information exchange between subsystems [9, 10]. As a result, they may fail to capture important dynamical features in regimes where interaction strengths are comparable and coherent evolution persists over microsecond timescales.

Similarly, electron–electron exchange and dipolar interactions are frequently omitted in order to preserve magnetic field sensitivity [4, 7, 8]. This assumption is problematic since both exchange and dipolar coupling strengths are distance dependent and, at $\mathrm{FAD}^{\cdot -}$–$\mathrm{TrpH}^{\cdot +}$ distances, are expected to be an order of magnitude larger than the Zeeman interaction [22, 26, 32]. One possible explanation, as proposed by Hore et al., is that the dipolar and exchange contributions may enter a cancellation regime, allowing the Zeeman interaction to perturb near-degenerate energy levels [26]. This work operates within such regimes

in order to preserve magnetosensitivity while retaining full interaction physics.

A central observable in theoretical treatments of the radical pair mechanism is the singlet yield, defined as the time-integrated probability of recombination through the singlet channel. This quantity has been widely adopted due to its direct connection to experimentally accessible reaction outcomes and its ability to encode magnetic field effects through variations in spin dynamics. As such, much of the RPM literature evaluates magnetosensitivity primarily through changes in singlet yield under varying magnetic field conditions.

While singlet yield provides a convenient and physically meaningful metric, it represents a coarse-grained characterization of the underlying quantum dynamics. In particular, it integrates over the full temporal evolution of the system, potentially obscuring transient coherent processes and frequency-dependent features arising from electron–nuclear interactions. These dynamical characteristics, including oscillatory behavior at or near the Zeeman frequency, are believed to play a central role in enabling magnetic field sensitivity.

1.4 Scope and Contributions of This Work

In this thesis, a fully quantum mechanical model of the radical pair system is developed in which both the electron spins and their nuclear environment are treated explicitly as a composite quantum system. The aim is to develop a fuller understanding of the RPM as a many-body quantum system, with particular emphasis on how its dynamics may be structured, controlled, and potentially engineered in biologically motivated regimes.

The first extension of this work is to develop a larger, more representative nuclear environment while staying fully quantum mechanical. The literature-common toy RPM models often include only one or two nuclei or rely on semiclassical approximations [4, 7, 8]. While recent work has been made in extending the complexity of the system certain limiting assumptions such as system symmetries or semiclassicality are employed [11, 22]. This work instead develops models with multiple nuclei coupled to each electron with biologically realistic coupling strengths spanning the full spectrum of anisotropic, coupling frequencies present in the system. These frequencies were inspired by density functional theory studies [33]. Although these nuclear environments are chosen to span biologically motivated coupling scales and anisotropies, they should be understood as structured toy models rather than molecule-specific microscopic reconstructions.

In contrast to standard approaches, environmental effects are incorporated through an open quantum system framework. Specifically, dissipation is modeled using a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation, where the dominant processes are taken to be spontaneous-emission–type transitions between system energy eigenstates [34]. Physically, these processes represent interactions with

the surrounding environment, including protein vibrational modes, solvent fluctuations, and other unresolved degrees of freedom. This provides a controlled and physically motivated baseline for assessing whether environmental coupling significantly influences radical pair dynamics.

A core objective of this work is to investigate the extent to which semiclassical approximations, such as the Overhauser field model, can reproduce the dynamics of the radical pair mechanism when compared to a full quantum mechanical treatment. While previous studies have largely focused on agreement at the level of singlet yields, this work expands the analysis to include time-resolved observables that capture the underlying dynamical structure of the system.

In particular, we introduce a systematic comparison between singlet populations, their running time averages, and final singlet yields across both semiclassical and fully quantum models. This approach allows us to distinguish between agreement in coarse-grained observables and agreement in the underlying spin dynamics. We demonstrate that, although semiclassical models can reproduce similar singlet yield values under certain conditions, they may fail to capture essential frequency-domain features associated with coherent electron–nuclear interactions.

By explicitly identifying where yield-based metrics succeed and where they obscure critical dynamical behavior, this work provides a more nuanced assessment of the validity of semiclassical descriptions. These results highlight the importance of retaining quantum coherence and structured frequency content when modelling magnetoreception, and motivate the use of fully quantum approaches in capturing biologically relevant spin dynamics.

Several additional physical mechanisms are explored in order to probe the structure and sensitivity of the dynamics. Exchange and dipolar interactions are explicitly included, with particular focus on operating within cancellation regimes that preserve magnetic field sensitivity while retaining full interaction physics. These interactions are of interest since they can create correlations between the electrons, allowing the flow of entanglement and quantum information and providing insight into the structure and chaos of entanglement dynamics.

The nuclear environment is further investigated as an active component of the system through the introduction of controlled perturbations in the form of nuclear spin flips. These operations are applied to both strongly coupled nuclei and weakly coupled nuclei. The strong coupling strengths being in the tens of MHz range whilst the weak couplings are around the unit MHz range [33]. Strongly coupled nuclei are expected to directly influence population dynamics, while weakly coupled nuclei primarily modify the spectral structure of the evolution. These tests provide insight into how different components of the nuclear environment contribute to and control the system dynamics.

Finally, a hypothesized nuclear decay process is introduced as a model for sudden environmental restructuring. In contrast to phenomenological noise models, this work develops a quantum-information-theoretic description of nuclear decay. Specifically, the β^- decay of a neutron is modelled. The products of the decay process are treated as three qubits, 1 qubit as the remaining proton and 2 emitted qubits as the electron and antineutrino. Adhering only to total spin conservation, a basis for the three-qubit space is built. An arbitrary superposition of these basis states can be used to create an arbitrary quantum channel taking the decaying neutron to the three product qubits. Tracing out the two emitted qubits simulates the emitting of the electron and antineutrino (assuming neither emitted particle is measured), resulting in a mixed state description of the remaining proton. This work then studies the impact of the decay on the electron spin states arising from entanglement between the remaining nuclear degrees of freedom and emitted particles. This allows decay to be embedded directly into the system dynamics as a localized, non-unitary operation, providing a controlled framework for studying how such processes induce decoherence and redistribute quantum correlations.

The overarching goal of this work is to determine the extent to which a fully quantum mechanical treatment is necessary to accurately describe radical pair dynamics, and to investigate how structured environments can influence and potentially control these dynamics. By grounding this analysis in biologically motivated models of the FAD–tryptophan radical pair, this thesis uses one of the best-studied RPM platforms as a realistic testbed for linking quantum information theory, many-body spin dynamics, and the prospective control and engineering of radical-pair systems.

CHAPTER 2

RADICAL PAIR MODEL AND HAMILTONIAN

This chapter defines the quantum mechanical model used to describe radical pair spin dynamics throughout this thesis. The radical pair is treated as a composite quantum system consisting of two interacting electron spins coupled to local nuclear spin environments. This model is extended in chapter 3 to include open-system dynamics. The formulation is quantum theory-based with chemical and biological contexts determining parameter choices and modeling assumptions, while the underlying dynamics are governed by a well-defined many-body Hamiltonian [8, 29].

This approach allows radical pair dynamics to be analyzed using tools from quantum information theory and open quantum systems, enabling the study of entanglement, correlations, and memory effects beyond traditional yield-based descriptions [23, 29]. The Hamiltonian introduced here provides the closed-system foundation to be extended in later chapters.

2.1 Hilbert Space and Degrees of Freedom

The system consists of two electron spins, labeled $e_1$and e_2, together with nuclear spin environments local to each electron. The total Hilbert space is

$$\mathcal{H} = \mathcal{H}_{e_1} \otimes \mathcal{H}_{e_2} \otimes \left(\bigotimes_{k=1}^{N_1} \mathcal{H}_{1k} \right) \otimes \left(\bigotimes_{k=1}^{N_2} \mathcal{H}_{2k} \right),$$

where $\mathcal{H}_{e_j} \cong \mathbb{C}^2$ denotes the Hilbert space of electron j, and $\mathcal{H}_{jk}$ denotes the Hilbert space of the k-th nucleus coupled to that electron.

Electron spins are modeled as spin-1/2 operators

$$\mathbf{S}_j = (S_{jx}, S_{jy}, S_{jz})$$

$$\mathbf{S}_{j\alpha} = \frac{1}{2}\sigma_{j\alpha}$$

with σ_α being the Pauli matrices. Nuclear spins are treated as either spin-1/2 or spin-1 degrees of freedom depending on isotope, allowing quadrupolar interactions to be included when $I \geq 1$ [23, 35].

All operators are understood to act on the full tensor-product Hilbert space by tensoring with identity operators on subsystems on which they do not act.

2.2 Singlet-Triplet Basis

The two-electron spin subspace $\mathcal{H}_{e_1} \otimes \mathcal{H}_{e_2}$may be conveniently decomposed into total-spin eigenstates. Defining the total electron spin operator

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2,$$

the joint electron Hilbert space splits into a one-dimensional singlet subspace with total spin $\mathbf{S} = 0$ and a three-dimensional triplet subspace with total spin $\mathbf{S} = 1$.

The singlet state is given by

$$|S\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right)$$

while the triplet manifold consists of the three orthonormal states

$$|T_{+1}\rangle = |\uparrow\uparrow\rangle$$

$$|T_0\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle \right)$$

$$|T_{-1}\rangle = |\downarrow\downarrow\rangle$$

These states form a complete basis for the electronic spin degrees of freedom and are eigenstates of both $\mathbf{S}_{tot}^2$ and $S_{tot,z}$. In the absence of interactions that couple different total-spin sectors, the singlet and triplet subspaces evolve independently. Many interactions relevant to radical pair dynamics act differently within this basis. Isotropic exchange interactions depend only on $\mathbf{S}_{tot}^2$ and therefore shift the singlet and triplet manifolds relative to one another, while leaving degeneracies within the triplet subspace intact [29]. In contrast, anisotropic interactions, such as hyperfine and dipolar couplings, do not generally commute with $\mathbf{S}_{tot}^2$ and can induce mixing between singlet and triplet states [4].

The singlet–triplet basis therefore provides a natural framework for analyzing radical pair spin dynamics. In particular, singlet–triplet energy splittings, degeneracies, and mixing mechanisms play a central role in determining coherent oscillations, sensitivity to weak magnetic fields, and the generation of quantum correlations [4, 8, 36]. These features become especially important when considering the combined effects of exchange and dipolar interactions, as discussed in subsequent sections.

During modelling the radical pair was initialized in the singlet state, consistent with the initial conditions of the magnetoreception radical pair[4, 36]. The nuclear spins were initialized in a maximally mixed state with certain investigations carried out on how dynamic changes within the nuclear environment affect the system's spin-dynamics [7].

2.3 Full Hamiltonian

The coherent dynamics of the radical pair are generated by a Hamiltonian of the form

$$H = H_Z + H_{hf} + H_{ex} + H_{dip} + H_Q.$$

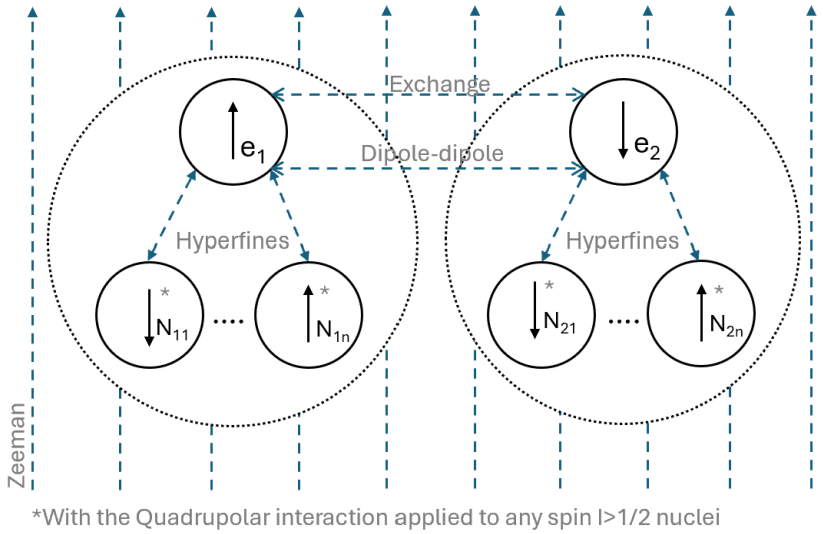


Figure 2.1 Schematic of the radical pair system and its associated interactions. Two electron spins (e_1, e_2) are coupled via exchange and dipole–dipole interactions, while each electron interacts with its local nuclear spin environment through anisotropic hyperfine couplings. An external magnetic field (Zeeman interaction) acts globally on all spins. For nuclei with spin $I > \frac{1}{2}$, quadrupolar interactions may also be present. This diagram summarizes all terms contributing to the full system Hamiltonian.

These terms describe electron Zeeman coupling to an external magnetic field, hyperfine interactions with surrounding nuclei, electron–electron exchange interaction, magnetic dipolar coupling, and nuclear quadrupole interactions, respectively. Figure 2.1 depicts a simplified diagram of the electrons, their nuclear environments and all interactions.

While simplified radical-pair models often neglect exchange and dipolar terms under assumptions of large electron separation or effective cancellation [23, 36], all interaction terms were retained in this work as independently tunable parameters. This allowed systematic exploration of how competing interaction mechanisms influence coherence, entanglement, and sensitivity to environmental structure.

2.4 Electron Zeeman Interaction

The interaction between the electron spins and an external magnetic field $\mathbf{B}$ is described by

$$H_Z = \gamma_e \, \mathbf{B} \cdot (\mathbf{S}_1 + \mathbf{S}_2),$$

where $\gamma_e = -g\mu_B$ is the electron gyromagnetic ratio.

The magnetic field defines a preferred quantization axis and breaks rotational symmetry. In radical pair systems relevant to biological settings, the field strength is typically weak compared to hyperfine energy scales, yet it plays a central role in modulating spin dynamics and singlet–triplet interconversion [4, 8, 36].

The nuclear Zeeman interaction is neglected due to the comparatively small nuclear gyromagnetic ratio, a standard approximation in radical-pair modeling [4, 37].

2.5 Hyperfine Interactions

Each electron spin interacts locally with a surrounding nuclear spin environment through hyperfine coupling. The hyperfine Hamiltonian is written in tensor form as

$$H_{hf} = \sum_{k=1}^{N_1} \mathbf{S}_1 \cdot \mathbf{A}_{1k} \cdot \mathbf{I}_{1k} + \sum_{k=1}^{N_2} \mathbf{S}_2 \cdot \mathbf{A}_{2k} \cdot \mathbf{I}_{2k},$$

where $\mathbf{A}_{jk}$is the hyperfine coupling tensor between electron j and nucleus k.

In the principal-axis frame, the coupling tensor is taken to be diagonal,

$$\mathbf{A}_{jk} = diag(A_x, A_y, A_z),$$

allowing for anisotropic hyperfine interactions. Such anisotropy is essential for generating singlet–triplet mixing and magnetic-field sensitivity [4, 22, 23, 32].

2.5.1 Overhauser Field Approximation

While the explicit hyperfine Hamiltonian provides a fully quantum mechanical description of the interaction between electron and nuclear spins, it is common in spin dynamics and radical pair modeling to employ a semiclassical approximation in which the nuclear environment is replaced by an effective magnetic field acting on the electron spins. This approach is known as the Overhauser field approximation [7].

Starting from the hyperfine Hamiltonian

$$H_{\mathrm{HF}} = \sum_k \mathbf{S} \cdot \mathbf{A}_k \cdot \mathbf{I}_k,$$

the nuclear spin operators may be replaced by their expectation values,

$$\mathbf{I}_k \to \langle \mathbf{I}_k \rangle,$$

yielding an effective magnetic field experienced by the electron spin,

$$\mathbf{B}_{\mathrm{OH}} = \frac{1}{\gamma_e} \sum_k \mathbf{A}_k \langle \mathbf{I}_k \rangle.$$

The hyperfine interaction then reduces to a Zeeman-like term

$$H_{\mathrm{OH}} = \gamma_e \mathbf{S} \cdot \mathbf{B}_{\mathrm{OH}}.$$

In this description, the discrete quantum nuclear environment is replaced by a classical stochastic magnetic field, often assumed to follow a Gaussian distribution when many weakly coupled nuclei contribute. The electron spin therefore evolves under an effective Hamiltonian consisting of the external magnetic field and the Overhauser field generated by the surrounding nuclei [7, 30].

Physically, the Overhauser field represents the mean magnetic field produced by the polarized nuclear environment. In systems with a large number of nuclei, the central limit theorem motivates modeling the field components as random variables whose variance depends on the hyperfine coupling strengths. This approximation significantly reduces computational complexity by removing the explicit nuclear degrees of freedom. Instead of simulating the full electron–nuclear Hilbert space, the electron spins evolve under a classical field sampled from an appropriate distribution. As a result, the Overhauser approximation is widely used in studies of electron spin decoherence and spin transport [7, 8, 27].

However, replacing the nuclear spins with a classical field removes several features that are intrinsic to the fully quantum description. The discrete dynamics of the nuclear spins are no longer resolved, electron–nuclear entanglement cannot develop, and the back-action of the electron subsystem on the nuclei is lost. Correlations between different nuclei are likewise absent. These omissions may be relatively harmless in large environments where coarse-grained field descriptions are adequate, but they can become significant in radical pair systems with finite nuclear baths and coherent electron–nuclear dynamics on biologically relevant timescales. For this reason, one objective of this thesis is to compare the full quantum nuclear model with the Overhauser-field approximation in order to determine under what conditions semiclassical descriptions fail to capture the dynamics of the radical pair system.

2.6 Electron-Electron Exchange Interaction

The isotropic exchange interaction between the two electrons is described by a Heisenberg coupling,

$$\hat{H}_J(r) = -J(r)\left(\hat{S}^2 - \hat{1}\right), \tag{2.1}$$

This interaction energetically splits the singlet and triplet manifolds and competes directly with hyperfine-induced singlet–triplet interconversion [4, 12, 26, 29].

Depending on the magnitude and sign of J, exchange interactions may suppress coherent oscillations or partially cancel other interaction terms, such as the magnetic dipolar coupling [26].

2.7 Magnetic Dipolar Interaction

The magnetic dipole–dipole interaction between the two electron spins arises from their mutual magnetic fields and, in its full form, is given by

$$H_{dip}^{full} = D\left[\mathbf{S}_1 \cdot \mathbf{S}_2 - 3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}})\right],$$

where $\hat{\mathbf{r}}$ is the unit vector connecting the two electron spins and D parameterizes the dipolar interaction strength.

In the presence of a static external magnetic field, and for field strengths large compared to transverse dipolar terms, the dipolar interaction may be simplified using the secular approximation. In this approximation, only terms that commute with the electron Zeeman Hamiltonian are retained, yielding an effective dipolar Hamiltonian of the form

$$\hat{H}_D(r) = D(r)\left[\hat{S}_z^2 - \frac{1}{3}\hat{S}^2\right], \tag{2.2}$$

up to a geometric factor determined by the orientation of the inter-radical axis relative to the magnetic field direction.

This secular form captures the dominant energetic splittings induced by the dipolar interaction while neglecting rapidly oscillating non-secular terms that average to zero in the rotating frame [8, 26, 37, 38]. The approximation is standard in radical pair theory and has been shown to reproduce the relevant singlet–triplet energy structure governing low-field spin dynamics [4, 26].

Throughout this work, the dipolar interaction was implemented in this secular form, with Dtreated as an effective control parameter incorporating both the magnitude of the dipolar coupling and its orientational dependence.

2.8 Exchange-Dipolar Cancellation

Early radical pair theory assumed that for spin dynamics to be susceptible to external magnetic fields like in avian magnetoreception, the exchange and dipolar interactions would need to be negligibly weak [7, 23, 27, 29]. Since both the dipolar and exchange interaction strengths depend on the separation distance between the constituents of the radical pair, the inter-radical pair distance was required to be too large to be compatible with the geometry of radical pairs such as flavin-tryptophan in cryptochrome. Work done by Efimova and Hore proved that, at certain distances, the dipolar and exchange interactions enter a cancellation regime. In this regime the energy-splitting caused by the dipolar and exchange interactions tends to zero allowing the Zeeman interaction to dominate. Equations (2.3) and (2.4) below express the decay with radical pair distance (r) of the dipolar and exchange coupling strengths respectively.

$$D(r) = \frac{3}{2}\frac{\mu_0}{4\pi}\frac{{g_e}^2\ {\mu_B}^2}{r^3} \tag{2.3}$$

$$J(r) = J_0\ e^{-br}, \quad b > 0 \tag{2.4}$$

At distances typical of the flavin-tryptophan radical pair (r = 2nm): D(r) ≈ -350uT and J(r) depending on differing Jo and b is also expected to be of the same order of magnitude as D(r) [26, 32].

By inspecting the matrix elements of $\hat{H}_J$ and $\hat{H}_D$ in the singlet–triplet basis, it becomes clear that an external magnetic field of order $50\,\mu$T, as in magnetoreception, would not generally dominate either the

dipolar or exchange effects [22, 26, 37]. Using the Efimova–Hore convention introduced in Eqs. (2.1) and (2.2), the exchange matrix elements are

$$\left\langle S\middle|\hat{H}_J\middle|S\right\rangle = J, \qquad \left\langle T_m\middle|\hat{H}_J\middle|T_m\right\rangle = -J, \qquad m = 0, \pm 1. \tag{2.5}$$

The corresponding secular dipolar matrix elements are

$$\left\langle S\middle|\hat{H}_D\middle|S\right\rangle = 0, \qquad \left\langle T_m\middle|\hat{H}_D\middle|T_m\right\rangle = D\left(m^2 - \frac{2}{3}\right), \qquad m = 0, \pm 1. \tag{2.6}$$

Thus, in the absence of Zeeman and hyperfine interactions, the combined exchange–dipolar energies are

$$E_S = J, \qquad E_{T_m} = -J + D\left(m^2 - \frac{2}{3}\right), \qquad m = 0, \pm 1. \tag{2.7}$$

A singlet–triplet degeneracy occurs when $E_S = E_{T_m}$, giving

$$J = -J + D\left(m^2 - \frac{2}{3}\right), \tag{2.8}$$

and therefore

$$D = \frac{2J}{m^2 - \frac{2}{3}}, \qquad m = 0, \pm 1. \tag{2.9}$$

This yields two commonly cited cancellation regimes:

$$|S\rangle\text{–}|T_0\rangle \text{ degeneracy}: \qquad m = 0 \quad \Rightarrow \quad D = -3J. \tag{2.10}$$

$$|S\rangle\text{–}|T_{\pm 1}\rangle \text{ degeneracy}: \qquad m = \pm 1 \quad \Rightarrow \quad D = 6J. \tag{2.11}$$

Since $D < 0$ for radical pairs in this convention, the $D = -3J$ regime corresponds to positive J, while the $D = 6J$ regime corresponds to negative J [26].

The different degeneracies introduced by each cancellation regime and their effect on spin dynamics and entanglement were investigated and are detailed in the results of this paper.

2.9 Nuclear Quadrupole Interaction

For nuclei with spin $I \geq 1$, electric quadrupole interactions are included through

$$H_Q = \sum_{(j,k)\in\mathcal{Q}} \mathbf{I}_{jk} \cdot \mathbf{Q}_{jk} \cdot \mathbf{I}_{jk},$$

where $\mathbf{Q}_{jk}$ is a traceless quadrupole tensor defining both the strength and orientation of the interaction.

Quadrupolar terms introduce additional anisotropy and internal nuclear dynamics, providing an additional mechanism through which nuclear degrees of freedom influence electron spin coherence [39].

2.10 Scope and Limitations

The Hamiltonian defined in this chapter describes coherent, closed-system spin dynamics. Environmental decoherence, dissipation, and non-Markovian effects are not included at this stage and are introduced in Chapter 3 using an open-quantum-system framework.

Throughout this thesis, units with $\hbar = 1$are used, and all parameters are chosen to span both biologically motivated regimes and idealized limits designed to probe fundamental dynamical behavior.

CHAPTER 3
OPEN QUANTUM SYSTEM FRAMEWORK

The Hamiltonian introduced in Chapter 2 describes the coherent spin dynamics of a radical pair and its surrounding nuclear environment under unitary time evolution. While this closed-system description is essential for identifying the intrinsic interaction mechanisms governing singlet–triplet interconversion, it is insufficient for modeling realistic biological systems.

In biological environments, radical pairs are embedded in complex protein scaffolds and solvent environments that contain a vast number of additional degrees of freedom. These environmental modes interact continuously with the electronic spins, leading to decoherence, dissipation, and relaxation processes that cannot be captured by unitary dynamics alone. As a result, a complete description of radical pair spin dynamics requires the framework of open quantum systems, in which the system of interest is coupled to an external environment (or bath) and evolves non-unitarily.

This chapter introduces the density matrix formalism and the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation used throughout this thesis as a baseline to model open-system effects. The physical interpretation of the dominant decoherence and relaxation mechanisms is discussed, along with the assumptions underlying Markovian bath models and their limitations in biological contexts[8, 25, 34].

3.1 Density Matrix Formalism

The state of an open quantum system is most naturally described using a density operator ρ, rather than a pure state vector. For a composite system consisting of a system S and an environment E, the total density operator evolves under unitary dynamics,

$$\rho_{SE}(t) = U(t)\ \rho_{SE}(0)\ U^{\dagger}(t),$$

where $U(t) = \exp(-iH_{SE}t)$and $H_{SE}$is the total Hamiltonian.

The reduced state of the system is obtained by tracing over environmental degrees of freedom,

$$\rho_S(t) = Tr_E\left[\rho_{SE}(t)\right].$$

Even when the combined system evolves unitarily, the reduced dynamics of $\rho_S(t)$ are generally non-unitary due to entanglement and information exchange between the system and its environment. This reduced description naturally captures loss of coherence, population relaxation, and irreversibility[25, 40, 41].

In the context of radical pair systems, the system density matrix $\rho(t)$ encodes not only population dynamics in the singlet and triplet manifolds but also quantum coherences and correlations, enabling direct

access to entanglement measures and information-theoretic quantities explored later in this thesis.

3.2 Markovian Master Equation

3.2.1 Born-Markov Approximation

To obtain a tractable equation of motion for $\rho_S(t)$, it is common to assume:

1. *Weak system–bath coupling* (Born approximation):
   The interaction between the radical pair spins and the environment is sufficiently weak that the bath remains near thermal equilibrium.

2. *Short bath correlation time* (Markov approximation):
   Environmental correlation functions decay much faster than the intrinsic system dynamics, allowing the bath to have no memory of past system states.

Under these assumptions, the reduced dynamics of the system become local in time and are governed by a Markovian master equation [25].

3.2.2 GKSL Master Equation

The most general form of a trace-preserving, completely positive, Markovian master equation is the GKSL equation [25, 42–44],

$$\frac{d\rho}{dt} = -i[H_S, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right),$$

Here, H_S denotes the system Hamiltonian introduced in Chapter 2, L_k are Lindblad or jump operators specifying the form of the system–environment coupling, and γ_k are the associated positive rates set by the bath spectral properties and temperature. The first term generates the coherent unitary evolution of the system, while the second term captures irreversible environmental effects such as decoherence, dephasing, and relaxation. This structure guarantees that the resulting dynamics remain completely positive and trace preserving, ensuring a physically consistent evolution of the density matrix. In this thesis, GKSL master equations are used as a controlled baseline for introducing environmental effects before considering more structured non-Markovian scenarios in later sections.

3.3 Interpretation of the Bosonic Bath

In this work, environmental effects are modeled using an effective bosonic bath. This bath does not represent a specific microscopic entity but rather an idealized description of the collective degrees of freedom external to the radical pair spin system.

Physically, the bosonic bath should be understood as an effective description of the many unresolved degrees of freedom surrounding the radical pair, including protein vibrational modes, solvent fluctuations, low-frequency collective motions of the molecular environment, and magnetic noise arising from nearby unresolved nuclear and electronic degrees of freedom. At physiological temperatures, these environmental modes are thermally populated and may be modeled as a continuum of harmonic oscillators [25, 30]. This picture provides a mathematically controlled way to incorporate decoherence and dissipation without explicitly simulating an intractably large environment.

3.3.1 Limitations of Markovian bath Models

While Markovian GKSL master equations are widely used and computationally efficient, they rely on strong physical assumptions that are not always well justified in biological settings. In particular, the Markovian approximation assumes that environmental memory is lost essentially instantaneously, preventing the buildup of sustained system–environment correlations and excluding the possibility of information backflow from the bath to the system. As a consequence, coherence and entanglement are forced to decay in a predominantly monotonic manner. Real biological environments, however, are often structured rather than featureless, with slow protein motions, discrete vibrational modes, and heterogeneous spin environments. Such structure can introduce memory effects and modify the spin dynamics in ways that lie beyond the reach of a purely Markovian description[25].

3.4 Transition to Non-Markovian Dynamics

Markovian baths provide a valuable baseline for understanding the dominant relaxation mechanisms acting on radical pairs. However, they inherently erase memory effects and constrain the dynamics to simple exponential decay [45, 46].

A central theme of this thesis is that structured environments can qualitatively modify radical pair dynamics, enabling non-Markovian behavior such as coherence revivals, entanglement backflow, and enhanced sensitivity to environmental control.

In subsequent chapters, the Markovian assumption is challenged by evidence for few-mode baths within the radical pair system. This introduces a non-Markovian regime to the model's dynamics. This transition enables systematic exploration of how environmental structure influences coherence, entanglement, and information flow in radical pair systems.

CHAPTER 4

QUANTIFYING QUANTUM CORRELATIONS AND OBSERVABLES

The radical pair mechanism has traditionally been characterized using chemically motivated observables, most notably singlet and triplet reaction yields. While such quantities are essential for connecting theory to experiment, they provide only a limited view of the underlying quantum dynamics. In particular, yield-based observables are insensitive to the structure and distribution of quantum correlations within the radical pair system and its surrounding nuclear environment.

In this thesis, the radical pair is treated as a composite open quantum system whose dynamics naturally generate entanglement, classical correlations, and information flow between subsystems. To characterize these effects, tools from quantum information theory are employed alongside conventional spin-chemical observables[4, 24]. This chapter introduces the subsystem partitions and correlation measure used throughout the remainder of the thesis and explains their physical relevance in the context of radical pair dynamics.

The central objective of this chapter is to define *what is measured* and *why these measurements matter*. By framing radical pair dynamics in quantum-information-theoretic language, this work provides a bridge between spin chemistry and modern many-body quantum theory.

4.1 Subsystem Decompositions

Quantum correlations depend fundamentally on how a system is partitioned [24]. Two distinct subsystem cuts are used throughout this work, as seen in Figure 4.1. Each cut probes a different physical aspect of the radical pair mechanism.

4.1.1 Electronic–Nuclear Partition:

($\mathbf{e_1}$,$\mathbf{e_2}$) | (nuclei)

The first partition separates the electronic spins from all nuclear degrees of freedom. This cut treats the radical pair electrons as the system of interest, with the nuclear spins acting as an internal environment.

This decomposition is particularly useful because it treats the electronic spins as the subsystem of interest and the nuclei as an internal environment. It therefore provides a natural framework for quantifying how entanglement is generated between electrons and nuclei, how hyperfine and quadrupolar interactions drive decoherence within the electronic sector, and how information leaks out of the electronic subsystem during the evolution. Because the nuclear degrees of freedom are retained explicitly rather than

traced out at the model level, this partition allows electron–nuclear entanglement to be studied directly instead of being absorbed into an effective phenomenological noise model.

4.1.2 Spatial Partition:

$(\mathbf{e_1}+\mathbf{n_1}) \mid (\mathbf{e_2}+\mathbf{n_2})$

The second partition groups each electron with its local nuclear environment, corresponding to a spatial separation between the two radicals.

This partition probes a different aspect of the dynamics by grouping each electron together with its local nuclear environment and then comparing the two radical-centered subsystems. In this way, it provides a measure of nonlocal correlation between the two spatially separated radicals, including correlations mediated indirectly through the nuclear degrees of freedom. It is therefore particularly useful for assessing whether the radical pair exhibits genuinely nonlocal quantum behavior, rather than dynamics that remain confined to local electron–nuclear clusters.

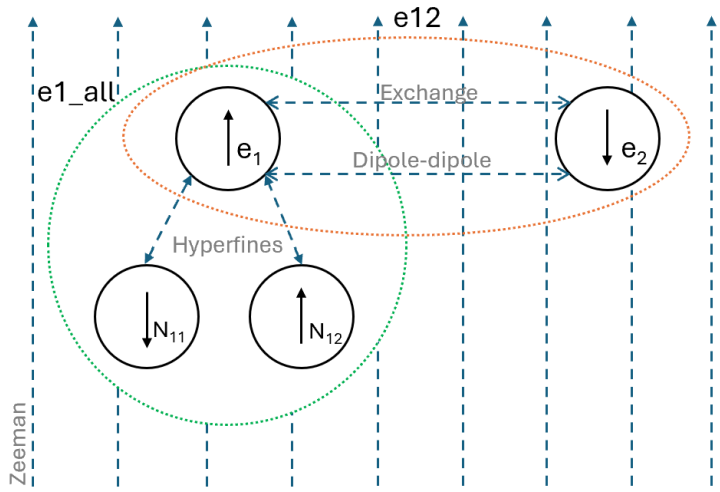


Figure 4.1 Illustration of subsystem partitions used for entanglement and correlation analysis. The green region (e_1 + local nuclei) represents a localized electron–nuclear subsystem, while the orange region ($e_1 + e_2$) represents the electronic subsystem. These bipartitions are used to compute von Neumann entropy enabling the study of quantum correlation flow between electrons and their nuclear environments.

4.2 Von Neumann Entropy

For a subsystem described by a reduced density matrix ρ_A, the von Neumann entropy is defined as

$$S(\rho_A) = -Tr(\rho_A \log \rho_A).$$

The von Neumann entropy quantifies the degree of mixedness of a subsystem and provides a measure of entanglement when the global state is pure [24]. In the context of open quantum systems, it also captures

entropy production due to environmental coupling [25].

In radical pair dynamics, increases in electronic entropy signal entanglement with nuclear spins or environmental modes [9]. While entropy alone does not distinguish classical from quantum correlations in mixed states, it provides a useful baseline diagnostic for information flow and decoherence [47].

4.3 Singlet Population and Time-Averaged Observables

In addition to quantum-information-theoretic measures, it is useful to track observables defined directly within the electronic spin subspace. The primary quantity considered in this work is the singlet population,

$$P_S(t) = \mathrm{Tr}\left[\rho(t)\,\Pi_S\right], \tag{4.1}$$

where $\Pi_S = |S\rangle\langle S|$ is the projector onto the electronic singlet state. This quantity represents the instantaneous overlap of the electronic subsystem with the singlet manifold and provides a direct measure of singlet–triplet interconversion.

The singlet population serves as a bridge between the microscopic quantum dynamics of the model and experimentally relevant spin observables [4, 29]. While it does not, in this work, correspond to a recombination yield, it retains the same structural role as in traditional radical-pair theory, allowing comparison with semiclassical and phenomenological models [31].

To characterize the longer-timescale behavior of the system and suppress rapid oscillatory structure, we also consider time-averaged forms of the singlet population. The running time average is defined as

$$\overline{P}_S(t) = \frac{1}{t}\int_0^t P_S(t')\,dt', \tag{4.2}$$

which provides a cumulative measure of singlet character up to time t.

In numerical simulations with discrete time steps, this quantity is evaluated as

$$\overline{P}_S(t_n) \approx \frac{1}{n}\sum_{k=1}^{n} P_S(t_k), \tag{4.3}$$

where t_n denotes the n-th time point.

The time-averaged singlet population acts as a coarse-grained observable that filters out high-frequency dynamics while retaining sensitivity to slower processes such as Zeeman-scale oscillations and interaction-induced envelope modulations. As such, it provides a complementary perspective to instantaneous observables and enables more direct comparison between different modeling approaches [4].

Importantly, agreement in time-averaged singlet population does not imply equivalence of the underlying dynamics. Distinct models may produce similar averaged behavior while differing significantly in their frequency content, coherence structure, and distribution of quantum correlations. For this reason, the singlet population and its time-averaged forms are interpreted throughout this work in conjunction

with entanglement and information-theoretic measures.

4.4 Reframing the Radical Pair Mechanism

By employing entanglement entropy alongside singlet yield, radical pair dynamics are reframed as a problem in quantum information flow rather than solely chemical kinetics.

This perspective reveals features—such as entanglement persistence, correlation redistribution, and non-Markovian revivals—that are invisible to yield-based analyses alone. It also allows direct comparison between radical pair systems and other mesoscopic quantum platforms, including spin chains and open quantum networks [48].

The tools introduced in this chapter form the quantitative backbone of the results presented in subsequent chapters, where they are used to diagnose environmental control, entanglement revival, and memory effects in radical pair systems.

CHAPTER 5

QUANTUM MODEL OF NUCLEAR DECAY

5.1 Motivation and Context

In addition to coherent Hamiltonian dynamics and environmental decoherence, nuclear processes may introduce discrete, non-unitary transformations within the radical pair system. While such processes are not typically included in standard radical pair models, they provide a useful theoretical framework for exploring how localized perturbations within the nuclear environment influence spin dynamics.

In this work, we introduce a hypothetical nuclear decay process acting on a single nucleus within the radical pair environment. The objective is not to model a specific microscopic decay channel in full detail, but rather to construct a minimal quantum-information-theoretic description of a decay event that: conserves spin-angular momentum, generates entanglement between the decay products, and introduces a nontrivial quantum channel on the remaining nuclear and electron spins.

This approach allows the decay process to be embedded directly into the radical pair simulation and analyzed alongside other mechanisms such as hyperfine interactions and controlled nuclear operations.

5.2 Spin-System Representation of β^- Decay

We model the decay process as a transition from an initial nuclear spin-1/2 degree of freedom to a three-particle product space consisting of the remaining proton together with the emitted electron and antineutrino [49]. Each of these product degrees of freedom is treated as a spin-1/2 system, so that all three are represented as qubits, while orbital angular momentum of the emitted particles is neglected. The decay therefore defines the mapping

$$H_{\text{in}} \cong \mathbb{C}^2, \qquad H_{\text{out}} = H_p \otimes H_e \otimes H_\nu \cong (\mathbb{C}^2)^{\otimes 3}.$$

Angular-momentum conservation is imposed by restricting the image of the decay map to the total-spin $J = 1/2$ subspace of the three-particle Hilbert space, matching the spin of the initial nuclear state [24].

5.3 Construction of the Decay Map

The decay process is defined as an isometric map

$$V : \mathcal{H}_{\text{in}} \to \mathcal{H}_p \otimes \mathcal{H}_e \otimes \mathcal{H}_\nu$$

$$V^\dagger V = I,$$

which embeds the initial nuclear state into the allowed total-spin subspace.

5.4 Lepton Pair Basis

The emitted lepton (electron–antineutrino) pair is first decomposed into singlet and triplet sectors:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1,$$

with basis states:

- singlet: $\mid s\rangle_{e\nu}$,
- triplets: $\mid t_+\rangle_{e\nu}, \mid t_0\rangle_{e\nu}, \mid t_-\rangle_{e\nu}$.

The total $J = 1/2$ subspace for the total space of the proton and leptons needs to be enforced. Coupling the proton spin with the lepton pair yields two distinct channels:

- *Singlet-lepton channel (A):* automatically total $J = 1/2$,
- *Triplet-lepton channel (B):* projected onto the $J = 1/2$ subspace via Clebsch–Gordan coefficients.

These define an orthonormal basis of the allowed output space [50, 51]

5.5 General Decay Mapping

The most general spin-conserving mapping takes the form

$$V \mid \pm\frac{1}{2}\rangle = a \mid A, \pm\frac{1}{2}\rangle + b \mid B, \pm\frac{1}{2}\rangle,$$

$$\mid a \mid^2 + \mid b \mid^2 = 1.$$

The coefficients a and b determine how the decay amplitude is distributed between the two angular-momentum channels.

The following assumptions about the model and decay process are used:

- Orbital angular momentum of decay products is neglected,
- The remaining proton retains the same hyperfine coupling tensor,
- The emitted electron and antineutrino are removed from the system,
- The decay occurs as a one-time, localized event.

These approximations allow the decay to be treated as a well-defined quantum channel embedded within the spin dynamics.

5.6 Reduced Dynamics of the Remaining Proton

After the decay, the emitted electron and antineutrino are not tracked as part of the radical pair system. The effective nuclear state is therefore obtained by tracing out these degrees of freedom:

$$\rho_p = Tr_{e\nu}\left[V\rho_{\text{in}}V^\dagger\right].$$

This induces a quantum channel acting on the remaining proton spin.

Defining

$$q \equiv \mid b \mid^2,$$

the reduced proton dynamics take the form of an SU(2)-covariant depolarizing channel:

$$\rho_p = (1-q)\rho_{\text{in}} + \frac{q}{3}\left(\sigma_x\rho_{\text{in}}\sigma_x + \sigma_y\rho_{\text{in}}\sigma_y + \sigma_z\rho_{\text{in}}\sigma_z\right),$$

5.7 Physical Interpretation

- $q = 0$: no entanglement with decay products, proton remains pure
- $0 < q < 1$: partial entanglement $\rightarrow$ partial depolarization
- $q \rightarrow 1$: maximal mixing within this model

Thus, the decay process acts as a *controlled decoherence mechanism* whose strength is directly tied to the entanglement generated during decay.

5.8 Embedding Decay into Radical Pair Model

To incorporate this process into the radical pair system, the decay is implemented as a single, instantaneous quantum operation acting on a selected nucleus at time t_{decay}.

The protocol is:

1. Evolve the full radical pair system under the Hamiltonian H(Chapter 2) up to t_{decay},
2. Apply the decay map V to the selected nuclear spin,
3. Trace out the emitted electron and antineutrino,
4. Continue unitary evolution with the updated nuclear state.

5.9 Scope and Interpretation

The decay model introduced here is deliberately phenomenological. It is designed to capture the essential quantum-structural features of nuclear decay—namely entanglement generation and induced decoherence—without attempting a full microscopic description of weak-interaction processes.

In particular, the parameter q is treated as a free quantity controlling the strength of entanglement generated between the remaining proton and the emitted leptons. The model does not attempt to resolve momentum, spatial degrees of freedom, or the detailed weak-interaction dynamics of the decay process. Instead, it should be understood as a hypothetical probe of how a localized nuclear event, through the generation of entanglement and subsequent tracing out of inaccessible degrees of freedom, could influence radical-pair spin dynamics. By varying q, the model provides a controlled way to explore how different levels of decay-induced entanglement propagate into the singlet–triplet dynamics and information flow of the larger spin system.

5.10 Connection to Observables

Although the emitted leptons are not directly observable within the radical pair system, their influence persists through the modified state of the remaining proton.

This motivates the use of standard observables–particularly singlet and triplet populations–as indirect probes of the decay process. Changes in these quantities following the decay event provide insight into how entanglement generated during decay propagates into the electronic spin dynamics.

This connection forms the basis for the analysis presented in Chapter 6.

CHAPTER 6

QUANTUM DYNAMICS, ENVIRONMENTAL STRUCTURE AND CONTROL IN THE RADICAL PAIR MODEL

Having established the quantum mechanical radical pair model and its open-system extension in the preceding chapters, we now investigate the dynamical behavior of the system through numerical simulation. The objective of this chapter is to determine how the interplay of coherent spin interactions and environmental coupling shapes the evolution of quantum correlations, information flow, and observable spin dynamics within the radical pair mechanism.

The radical pair is modeled as a composite many-body quantum system consisting of two electron spins coupled to surrounding nuclear environments through anisotropic hyperfine interactions. As discussed in Chapter 2, the coherent dynamics of this system arise from a Hamiltonian containing Zeeman, hyperfine, exchange, dipolar, and quadrupolar terms. These interactions generate singlet–triplet interconversion, electron–nuclear entanglement, and complex multi-spin dynamics that cannot generally be captured by semiclassical approximations.

In realistic biological environments, however, the radical pair is not an isolated system. Protein vibrations, solvent fluctuations, and unresolved environmental degrees of freedom introduce decoherence and relaxation processes that can modify spin dynamics. Chapter 3 introduced the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation as a baseline framework for incorporating such environmental effects in a controlled manner. Within this framework, dissipative and dephasing processes are modeled through Lindblad jump operators that couple the radical pair to an effective bosonic bath representing its surrounding molecular environment.

The results presented in this chapter pursue three central objectives. First, we investigate how rapidly open-system dynamics drive the radical pair toward a steady state and quantify the timescale over which the system retains memory of its initial condition. Second, we compare the full quantum nuclear model with semiclassical approximations of the nuclear environment to identify regimes where explicitly quantum nuclear dynamics become essential. Finally, we seek to characterize and test our fully quantum-mechanical model of the radical pair mechanism. This involves tracking entanglement within the system and modeling how changes to the nuclear environment affect the spin dynamics of the model.

Together, these investigations establish the dynamical regimes in which a fully quantum mechanical treatment of the radical pair is necessary and identify the physical mechanisms responsible for long-lived coherence, correlation exchange, and environmental memory effects.

6.1 Open-System Dynamics and Thermalization

Radical pair reactions occur within complex biological environments containing many uncontrolled degrees of freedom, including protein vibrations, solvent fluctuations, and unresolved nuclear spins. As a result, the radical pair spin system cannot be considered perfectly isolated and must, in principle, be treated as an open quantum system. While the Hamiltonian dynamics introduced in Chapter 2 capture the intrinsic spin interactions responsible for singlet–triplet interconversion, environmental couplings may introduce additional decoherence and dissipative processes that modify the effective spin dynamics.

Beyond simple decoherence, environmental interactions may qualitatively alter system behavior through quantum measurement effects. In particular, Kominis proposed that spin-selective recombination channels in radical pair reactions effectively perform continuous measurements of the singlet projector, potentially driving the system into a quantum Zeno regime in which specific spin subspaces become dynamically stabilized. In this regime, sufficiently strong environmental coupling suppresses coherent transitions and confines the system to restricted regions of Hilbert space.

Motivated by this possibility, we investigate whether biologically inspired environmental dissipation could produce analogous effects within a more general open-system description of the radical pair. Specifically, we explore whether spontaneous-emission–type Lindblad processes acting on the system's energy eigenbasis could generate Zeno-like stabilization or protected dynamical subspaces within the radical pair Hilbert space.

6.1.1 System Setup

Simulating open-system radical pair dynamics over the long timescales required for thermalization can be computationally demanding, particularly when multiple nuclear spins are included explicitly. To make these simulations tractable while retaining the essential features of the radical pair mechanism, a simplified model was employed for the thermalization investigations. The system consists of two electron spins forming a radical pair. Electron 1 is coupled to two nuclear spins through anisotropic hyperfine interactions, while electron 2 is assumed to have no local nuclear environment. The two nuclei represent a minimal internal environment containing both strong and weak hyperfine couplings. The hyperfine tensors used in these simulations correspond to one strongly coupled nucleus and one weakly coupled nucleus, $A_{\text{strong}} \sim 20$ MHz and $A_{\text{weak}} \sim 20$ kHz, providing a minimal representation of the heterogeneous hyperfine environments expected in realistic radical pair systems. The external magnetic field was fixed at approximately Earth strength, $B \approx 47\ \mu$T, aligned along the z-axis. Electron–electron exchange and dipolar interactions were included using the cancellation regime $D = -3J$, consistent with the parameter regime discussed in Chapter 2. This regime restores degeneracies within the singlet–triplet manifold that

allow the weak Zeeman interaction to play a dominant role in the spin dynamics.

Open-system effects were incorporated using the GKSL master equation introduced in Chapter 3,

$$\dot{\rho} = -i[H, \rho] + \mathcal{L}(\rho),$$

where the dissipator

$$\mathcal{L}(\rho) = \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

models environmental decoherence and dissipation. The Lindblad operators were chosen to represent spontaneous-emission–like transitions between energy eigenstates of the system Hamiltonian. The dissipation rates used in these simulations span the range $\gamma_k \sim 10^{-9}\,\mathrm{s}^{-1} - 10^{-3}\,\mathrm{s}^{-1}$, consistent with estimates for weak environmental coupling in biological systems.

6.1.2 Thermalization Timescales

To quantify how environmental coupling drives the system toward equilibrium, the approach to a steady state was analyzed using the trace distance between the instantaneous system state and a late-time steady-state proxy. The steady state was estimated by averaging the density matrix over the final portion of the simulation time window and comparing this average state with the evolving system state. The resulting trace distance provides a measure of how much information about the initial state remains in the system at a given time.

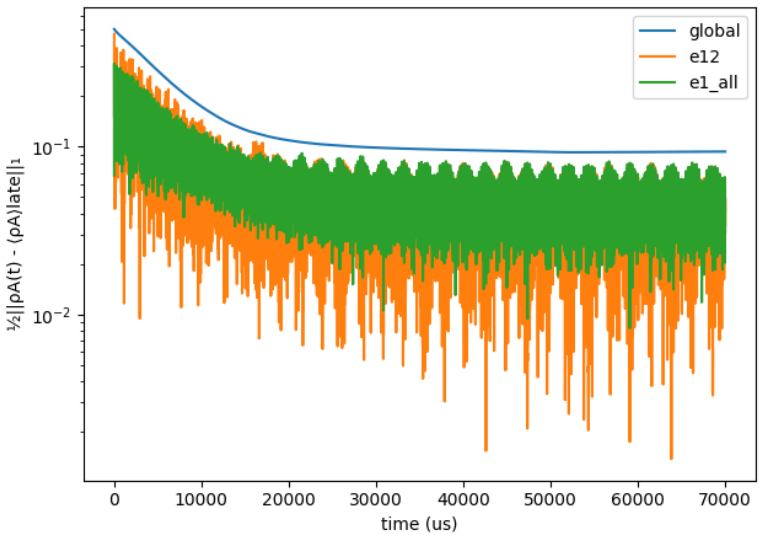


Figure 6.1 Open-system trace distance to a late-time steady-state proxy versus time. The decay is monotonic, but the relaxation timescales remain long compared with the microsecond radical-pair lifetime.

Figure 6.1 shows the decay of the trace distance to the late-time steady-state proxy. The results reveal a clear separation between the timescales governing coherent spin dynamics and those associated with environmental relaxation. While coherent oscillations driven by hyperfine and Zeeman interactions occur on microsecond timescales, the open-system dynamics approach a steady state only on much longer timescales of order milliseconds. This behavior indicates that the radical pair system retains a substantial memory of its initial condition over the entire microsecond lifetime typically associated with radical pair recombination.

6.1.3 Dependence on Dissipation Rates

To determine whether stronger environmental coupling could influence the dynamics on biologically relevant timescales, the GKSL dissipation rates were systematically increased across several orders of magnitude. To ensure the open-system Lindbladian stayed consistent with the weak-coupling Born approximation, the common convention of keeping the maximum dissipation rates, $\gamma_{D,\max}$, at most ten times less than the characteristic frequencies of the system, was followed. The characteristic frequency was chosen to be the Zeeman frequency, f_z, where $f_z \approx 1.4$ MHz $\Rightarrow \omega_Z \approx 8.8$ rad/μs. Defining $\epsilon = \gamma_D/\omega_z$, multiple values of ϵ were used to capture when GKSL bath effects become non-negligible on microsecond timescales.

The behavior across the rate sweep is summarized in Figure 6.2. The results demonstrate that the dissipation rates expected from spontaneous-emission–type processes are far too small to affect the system on microsecond timescales. For the physically motivated parameter regime, the open-system dynamics are essentially indistinguishable from the purely unitary evolution of the closed system. Only when the dissipation rates are artificially increased by several orders of magnitude do environmental effects begin to visibly modify the spin dynamics within the radical pair lifetime.

This behavior can be understood by comparing the characteristic scales of coherent and dissipative processes. The dominant coherent frequencies in the system are set by the Zeeman and hyperfine interactions, which lie in the megahertz regime. In contrast, biologically motivated dissipation rates are many orders of magnitude smaller, preventing the environment from significantly influencing the dynamics over microsecond timescales.

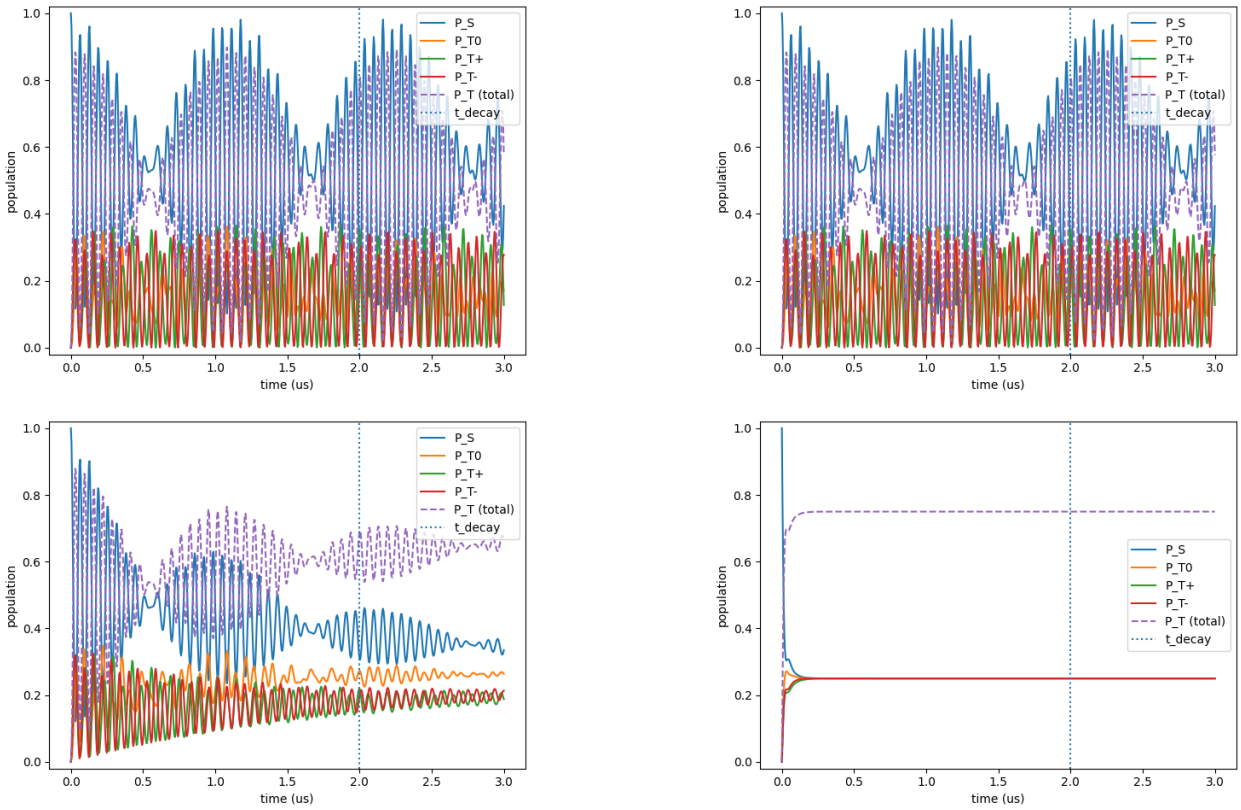


Figure 6.2 Singlet and triplet dynamics under increasing GKSL rates. The physically motivated rates are too small to alter the microsecond-scale dynamics appreciably; only strongly exaggerated rates visibly modify the evolution.

6.1.4 Implications for Zeno Stabilization

One of the motivations for introducing open-system dynamics was to explore whether environmental coupling could produce quantum Zeno–like stabilization of particular spin subspaces. In principle, sufficiently strong dissipative channels can act as continuous measurements, suppressing coherent transitions and confining the system to restricted regions of Hilbert space. However, the simulations presented here indicate that the dissipation rates required to induce such Zeno stabilization are far larger than those expected for biological environments. For realistic environmental coupling strengths, the radical pair dynamics remain effectively unitary on magnetoreception timescales. Consequently, spontaneous-emission–type relaxation processes are unlikely to play a significant role in shaping radical pair spin dynamics or generating Zeno-protected subspaces in the system's energy eigenbasis.

6.1.5 Summary of Open-System Results

The open-system simulations reveal that the biologically plausible environmental dissipation employed in this paper is too weak to significantly alter radical pair dynamics within the microsecond lifetime relevant to magnetoreception. Thermalization occurs only on millisecond timescales, far beyond the duration over which coherent spin dynamics influence the chemical recombination process.

This result implies that spontaneous-emission–type environmental relaxation is unlikely to provide a mechanism for quantum Zeno stabilization in radical pair systems. Instead, the dominant features of the dynamics are governed by the intrinsic Hamiltonian interactions between electron and nuclear spins.

At the same time, the long thermalization times observed in these simulations highlight the importance of internal nuclear environments as potential sources of effective memory and information storage within the radical pair system.

These findings motivate the subsequent investigations in this thesis, which explore environmental structures that go beyond simple Markovian GKSL descriptions. In particular, structured or non-Markovian environments may enable information backflow, coherence revivals, and entanglement redistribution that cannot be captured within the Markovian framework considered here.

The following sections therefore investigate mechanisms through which internal nuclear dynamics and structured environments can generate non-Markovian behavior and long-lived quantum correlations in radical pair systems

6.2 Overhauser Field Approximation vs Full Quantum Nuclear Dynamics

A common simplification in radical-pair simulations is to replace the discrete quantum nuclear spin environment with an effective classical magnetic field, known as the Overhauser field. In this approximation, the nuclei are treated as a stochastic field acting on the electron spins, thereby neglecting quantum correlations, entanglement, and dynamical back-action between the electrons and nuclei. While computationally efficient, the validity of this approximation in regimes relevant to magnetoreception remains unclear.

6.2.1 Model Setup

For these simulations, the radical pair model was extended beyond the simplified thermalization system to include a larger nuclear environment. Each electron spin was coupled to three nuclear spins, yielding a total of six nuclear spins in the system. The hyperfine coupling structure was chosen to mimic heterogeneous nuclear environments, with each electron coupled to one strongly coupled nucleus, one moderately coupled nucleus, and one weakly coupled nucleus. The hyperfine tensors were chosen to produce

anisotropic interactions and were inspired by hyperfine couplings calculated for flavin adenine dinucleotide (FAD) and tryptophan radical (Trp) molecules using density functional theory in previous studies [22]. All simulations in this section were performed under closed-system dynamics, isolating the effect of the nuclear environment itself rather than environmental dissipation. The exchange and dipolar parameters were chosen in the $D = -3J$ cancellation regime discussed earlier, restoring degeneracies in the singlet–triplet manifold and allowing the Zeeman interaction to play a dominant role in the spin dynamics. All simulations are performed with the nuclei initialized in a maximally mixed state, ensuring that no artificial polarization or symmetry breaking is introduced. Within the Overhauser approximation, the nuclear spins were replaced by classical effective magnetic fields acting on each electron. These fields were generated by sampling nuclear spin configurations and projecting their hyperfine interactions into an effective field acting on the electrons. Two complementary scenarios are considered: (i) an ensemble-averaged study over many Overhauser realizations, and (ii) a single-shot realization to probe the underlying dynamics that may be obscured by averaging. All simulations in this section were performed under closed-system dynamics, isolating the effect of the nuclear environment itself rather than environmental dissipation

6.2.2 Ensemble-averaged Dynamics

We first consider the ensemble-averaged behavior of the system. Figure 6.3 shows the singlet population $P_S(t)$ for both the full quantum model and the Overhauser approximation. In both cases, the system rapidly departs from the initial singlet state and settles into fluctuations around a quasi-stationary value. At first glance, the agreement between the two models appears reasonable, with both exhibiting similar overall trends.

However, more sensitive diagnostics reveal systematic discrepancies. The running time-averaged singlet population,

$$\overline{P}_S(t) = \frac{1}{t} \int_0^t P_S(t')\, dt', \tag{6.1}$$

is shown in Figure 6.4. While both models converge toward steady values, the Overhauser approximation consistently underestimates the long-time average relative to the full quantum result.

This discrepancy is further quantified in Figure 6.5, where the final time-averaged values are compared. Although the difference is modest, it is consistent and indicates that the Overhauser approximation does not fully capture the steady-state behavior of the quantum system.

A more pronounced distinction emerges in the frequency domain. Figure 6.6 shows the Fourier transform of $P_S(t)$. The full quantum model exhibits a structured spectrum with identifiable peaks across a range of frequencies, including components near the Zeeman scale. In contrast, the Overhauser

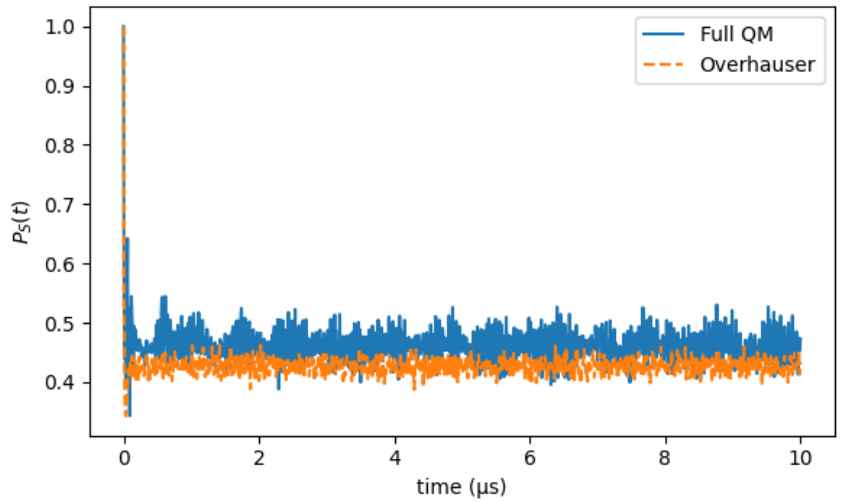


Figure 6.3 Singlet population $P_S(t)$ for the ensemble-averaged maximally mixed nuclear state. The Overhauser approximation reproduces the general decay and fluctuation scale but exhibits reduced structure compared to the full quantum model.

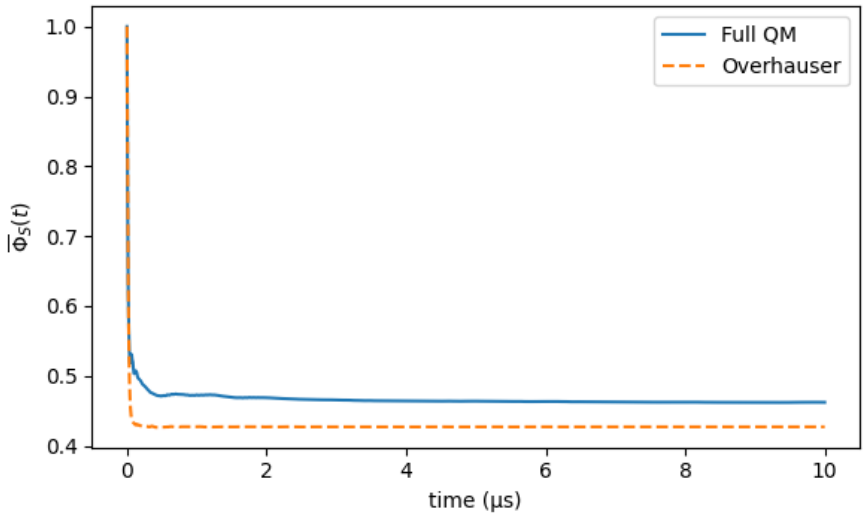


Figure 6.4 Running time-averaged singlet population $\overline{P}_S(t)$ for the ensemble study. The Overhauser approximation converges to a systematically lower value than the full quantum model.

approximation produces a significantly flatter spectrum with reduced structure, indicating a loss of coherent frequency content.

6.2.3 Single-Shot Dynamics

To probe the origin of these discrepancies, we next consider a single realization of the Overhauser field. The resulting singlet population dynamics are shown in Figure 6.7.

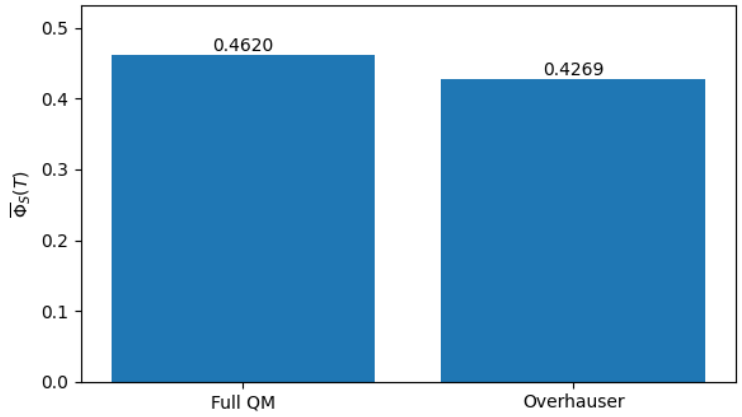


Figure 6.5 Final time-averaged singlet population for the ensemble study. The Overhauser approximation underestimates the long-time average relative to the full quantum model.

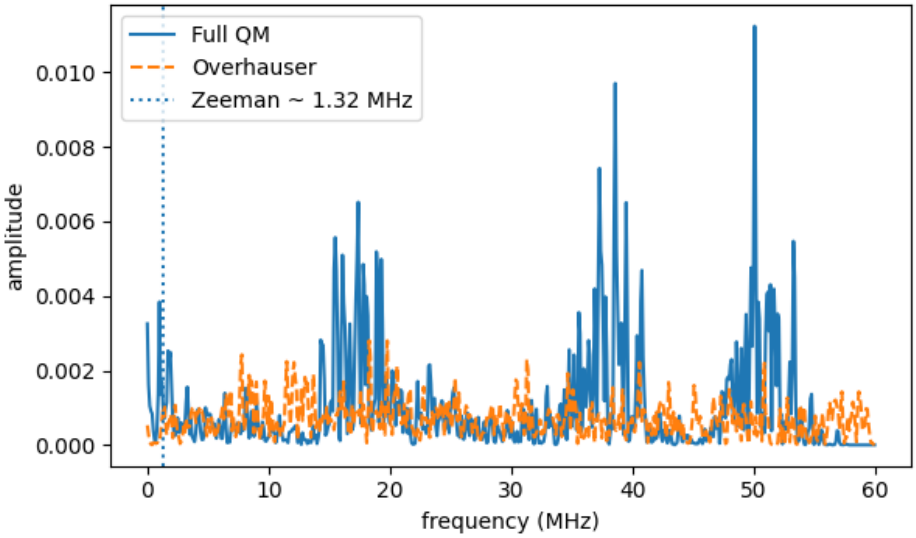


Figure 6.6 Fourier transform of the singlet population for the ensemble study. The full quantum model exhibits structured frequency components, while the Overhauser approximation produces a flatter, less coherent spectrum. The dashed line indicates the Zeeman frequency scale.

In contrast to the ensemble-averaged case, the Overhauser trajectory now exhibits large-amplitude, highly irregular oscillations spanning a wide range of values. This behavior is markedly different from the relatively stable fluctuations observed in the full quantum model, highlighting the intrinsically stochastic nature of the semiclassical approximation.

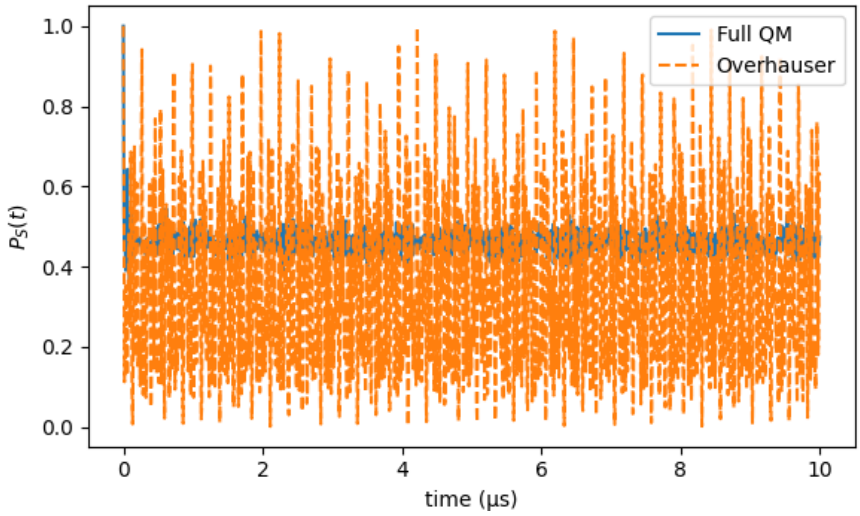


Figure 6.7 Singlet population $P_S(t)$ for a single Overhauser realization. The semiclassical trajectory exhibits large-amplitude, irregular oscillations compared to the more constrained quantum dynamics.

The running time-averaged behavior, shown in Figure 6.8, again reveals a systematic bias. While both models approach steady values, the Overhauser result converges to a significantly lower value than the full quantum model, reinforcing the discrepancy observed in the ensemble study.

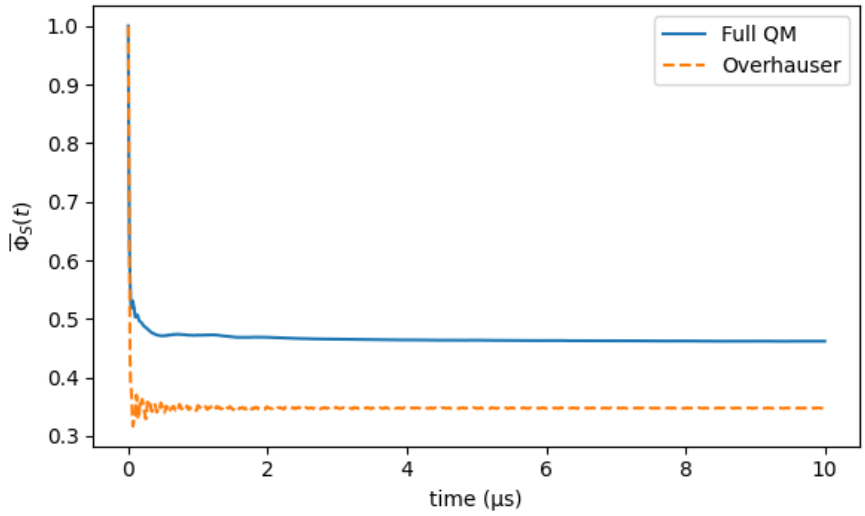


Figure 6.8 Running time-averaged singlet population for a single Overhauser realization. The Overhauser approximation converges to a lower steady value, indicating a systematic bias.

This difference is further reflected in the final time-averaged values shown in Figure 6.9. Notably, the apparent agreement seen in the ensemble case arises from averaging over many such irregular trajectories rather than from an accurate representation of the underlying dynamics.

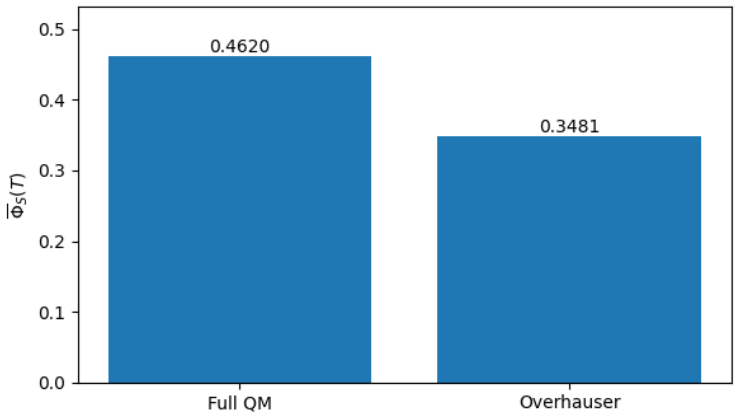


Figure 6.9 Final time-averaged singlet population for the single-shot study. The deviation from the full quantum result is significantly larger than in the ensemble case.

Finally, the Fourier spectra in Figure 6.10 show that the Overhauser approximation produces sharp, high-amplitude peaks distributed across a broad frequency range. This indicates incoherent mixing of frequencies introduced by the stochastic field. In contrast, the full quantum model retains a more structured spectrum with identifiable low-frequency components.

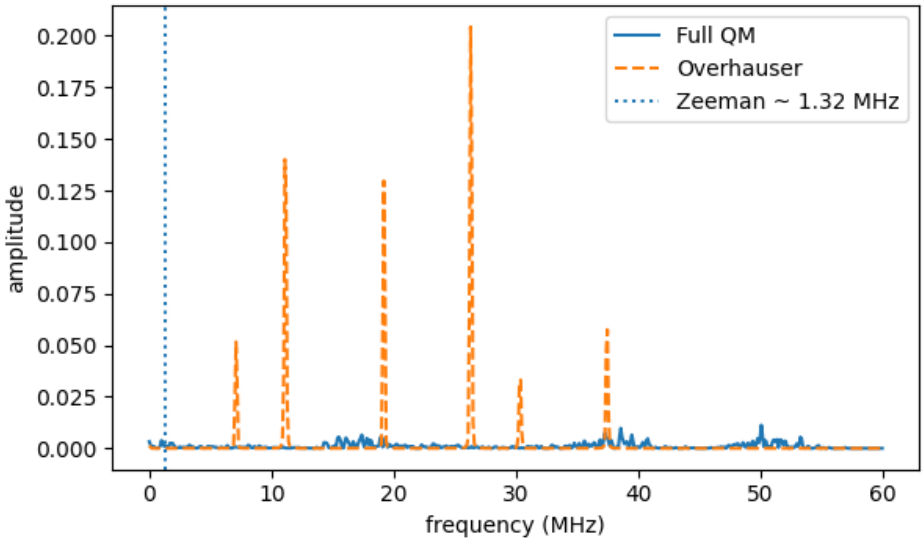


Figure 6.10 Fourier transform of the singlet population for a single Overhauser realization. The Overhauser spectrum is dominated by sharp, incoherent peaks, while the full quantum model retains structured, physically meaningful frequency components.

Table 6.1 Hyperfine robustness sweep for the full quantum and Overhauser ensemble models. All hyperfine tensors were scaled uniformly by factors of 0.8, 1.0, and 1.2 relative to the baseline values, while all other parameters were held fixed. The simulations were performed for a fixed $5\,\mu$s runtime at $B = 0.05\,\mathrm{mT}$ in the $D = -3J$ cancellation regime with 400 Overhauser samples. Shown are the final time-averaged singlet yields and the trace/FFT RMS differences relative to the baseline ($1.0\times$) case.

Scale	$\bar{\Phi}_S^{\mathrm{QM}}(T)$	$\bar{\Phi}_S^{\mathrm{OH}}(T)$	QM Trace RMS	OH Trace RMS	QM FFT RMS	OH FFT RMS
$0.8\times$	0.476927	0.440468	0.039987	0.022569	0.055070	0.038027
$1.0\times$	0.463583	0.426964	0.000000	0.000000	0.000000	0.000000
$1.2\times$	0.451502	0.417872	0.038664	0.019805	0.054330	0.038747

6.2.4 Model Robustness

A hyperfine robustness sweep was performed by uniformly scaling all hyperfine tensors by factors of 0.8, 1.0, and 1.2 relative to the baseline values, while holding all other parameters fixed. A summary can be seen in Table 6.1. In both the full quantum and Overhauser models, the final time-averaged singlet yield decreased systematically as the hyperfine scale increased. The Overhauser approximation remained consistently lower than the full quantum model at all three scales, indicating that its bias is not restricted to a single parameter choice. Although both the time-domain traces and Fourier spectra changed under the perturbation, the qualitative behavior remained stable, with the full quantum model exhibiting greater sensitivity in its spectral structure. These results show that the main discrepancy between the Overhauser and full quantum descriptions persists across a reasonable neighborhood of hyperfine strengths, supporting the conclusion that the baseline behavior is not anecdotal.

Table 6.2 Field-angle robustness sweep for the full quantum and Overhauser ensemble models. The external magnetic field of fixed magnitude $B_0 = 0.05\,\mathrm{mT}$ was rotated within the xz-plane through angles of 0°, 30°, 60°, and 90° relative to the z-axis. Shown are the trace and FFT RMS differences relative to the baseline 0° case.

Angle	QM Trace RMS	OH Trace RMS	QM FFT RMS	OH FFT RMS
0°	0.000000	0.000000	0.000000	0.000000
30°	0.017569	0.016398	0.019636	0.030822
60°	0.027092	0.016034	0.026085	0.034270
90°	0.032928	0.015972	0.027094	0.034838

A field-angle robustness sweep was performed by rotating the external magnetic field within the xz-plane while keeping its magnitude fixed at $B_0 = 0.05\,\mathrm{mT}$. Simulations were carried out for field angles of 0°, 30°, 60°, and 90° relative to the z-axis, with all other parameters held fixed. The resulting singlet-population traces remained within the same broad dynamical regime across the sweep, but showed clear orientation-dependent variation in their detailed time-domain structure. In the full quantum model, the trace RMS difference relative to the 0° baseline increased from 1.76×10^{-2} at 30° to 3.29×10^{-2} at

90°, indicating a progressively stronger dynamical response as the field was rotated away from the baseline direction. The Overhauser traces also varied with angle, but more weakly, with RMS differences remaining near 1.6×10^{-2} across the sweep. The Fourier spectra exhibited similarly clear angular dependence, with the full quantum FFT RMS difference increasing to 2.71×10^{-2} at 90°, while the Overhauser spectra showed broader but less structured variation, with FFT RMS differences reaching 3.48×10^{-2}. These results show that the baseline behavior is not tied to a single field orientation and that the angular response is carried more clearly in the detailed dynamical and spectral structure than in the final time-averaged singlet yield.

6.2.5 Discussion and Implications

Taken together, these results demonstrate that the Overhauser field approximation fails to reproduce key dynamical features of the full quantum model. While ensemble averaging can produce superficially similar time-domain behavior, it obscures significant discrepancies in both the frequency content and statistical structure of the dynamics.

The origin of this breakdown lies in the neglect of quantum correlations and dynamical back-action between the electrons and nuclei. In the full quantum model, the nuclear spins form a structured, non-Markovian environment that exchanges information with the electron subsystem, leading to coherent frequency mixing and constrained dynamics. The Overhauser approximation replaces this structured environment with a stochastic field, eliminating these correlations.

This distinction has direct implications for magnetoreception. The sensitivity of the radical-pair mechanism depends on well-defined low-frequency (Zeeman-scale) dynamics. The suppression and distortion of these features in the Overhauser approximation suggest that it may fail to capture the mechanisms responsible for magnetic-field sensitivity, even when the nuclear ensemble is maximally mixed.

These findings therefore highlight the necessity of a fully quantum mechanical treatment of the nuclear environment when accurate dynamical predictions are required.

6.3 Entanglement Dynamics and Exchange–Dipolar Interaction Effects

While the singlet–triplet populations of radical pair systems provide a direct observable for magnetoreception, they capture only a limited portion of the underlying quantum dynamics. The full radical pair system consists of a network of interacting spins in which quantum correlations can flow between electrons and nuclei. Tracking these correlations provides insight into how quantum information propagates through the system and whether the nuclear environment acts merely as classical noise or as an active participant in the spin dynamics. To investigate this behavior, the von Neumann entropy of various

subsystems was tracked during closed-system evolution.

This analysis serves two purposes within the broader goal of validating the need for a full quantum mechanical treatment of radical pair dynamics. First, it reveals whether the nuclear environment behaves as a passive background field or as a dynamical quantum reservoir capable of storing and returning quantum information. Second, it allows us to determine how additional interaction terms, particularly exchange and dipolar couplings, influence the flow of entanglement within the radical pair system

6.3.1 Model setup

To isolate the phenomenology of entanglement exchange, a simplified radical pair model was used for these simulations. The system consists of two electron spins forming the radical pair, with two nuclear spins coupled to electron 1 and no nuclei coupled to electron 2. This minimal configuration preserves the essential structure of electron–nuclear interactions while keeping the Hilbert space small enough to clearly visualize subsystem dynamics. The hyperfine couplings were chosen to include both a strongly coupled nucleus and a weakly coupled nucleus, producing two distinct interaction scales. As in the previous sections, the external magnetic field was fixed near Earth strength ($B \approx 47$ uT), corresponding to a Zeeman frequency of approximately $f_z \approx 1.4$ MHz. All simulations presented in this section were performed under closed-system dynamics, allowing the intrinsic Hamiltonian interactions alone to determine the evolution of correlations within the system. This simplified model was chosen intentionally since the goal is to reveal general qualitative features of entanglement flow rather than reproduce a specific molecular system. The reduced nuclear environment provides a clearer view of the underlying dynamical mechanisms.

The entanglement dynamics were analyzed by computing the von Neumann entropy

$$S(\rho_A) = -\mathrm{Tr}(\rho_A \log_2 \rho_A)$$

for several subsystems obtained by tracing out the remaining degrees of freedom.

Three subsystem partitions were examined:

- the global electronic subsystem, consisting of both electrons,
- the electron–nuclear subsystem containing electron 1 and its nuclei,
- the total system.

Because the overall system evolves unitarily, the total entropy of the full system remains constant. However, entropy within subsystems can fluctuate as entanglement is redistributed among different degrees of freedom.

To determine the role of electron–electron couplings in shaping entanglement dynamics, simulations were performed both with and without the exchange and dipolar interaction terms.

6.3.2 Effect of Exchange and Dipolar Couplings

We now investigate the role of exchange (J) and dipolar (D) interactions in shaping both the entanglement structure and observable singlet dynamics of the radical pair system. These interactions introduce direct electron–electron coupling, fundamentally altering the pathways through which quantum correlations are generated and redistributed between the electrons and the nuclear environment.

6.3.3 Subsystem entropy and entanglement structure

We begin by examining the subsystem von Neumann entropies for the closed system evolution, shown in Figures Figure 6.11 and Figure 6.12. Three bipartitions are considered: the global entropy, the electron pair subsystem (e_1, e_2), and a mixed partition (e_1 + nuclei).

In the absence of exchange and dipolar interactions (Figure 6.11), the entropy dynamics exhibit highly regular, almost periodic behavior. The (e_1, e_2) subsystem entropy oscillates rapidly between low and high values, indicating repeated cycles of entanglement generation and decay between the electrons and the nuclear environment. Meanwhile, the (e_1 + nuclei) partition remains essentially constant, suggesting that correlations are largely confined to localized electron–nuclear interactions with minimal global redistribution.

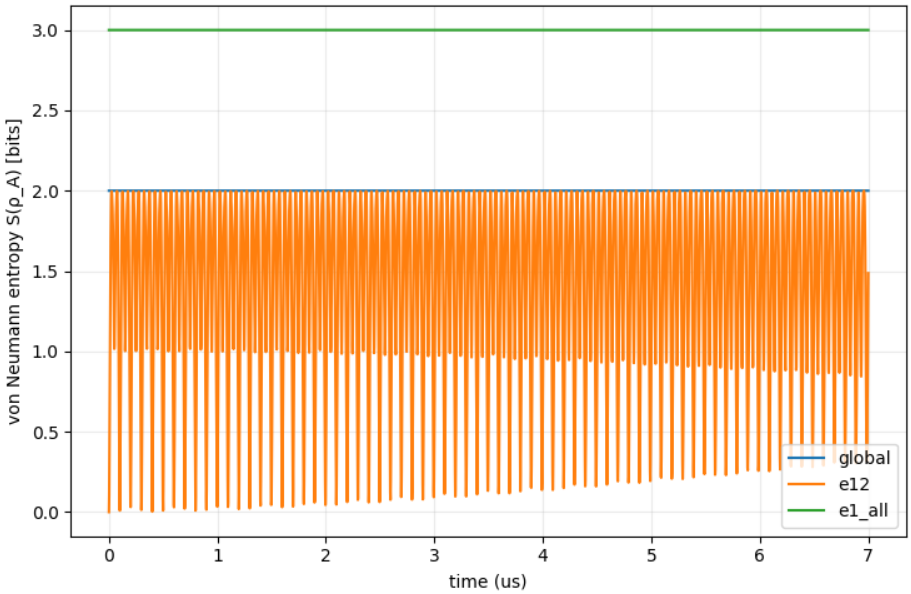


Figure 6.11 Subsystem von Neumann entropies without exchange and dipolar interactions. The dynamics are highly regular, with oscillatory entanglement exchange primarily between individual electrons and their local nuclear environments.

In contrast, when exchange and dipolar interactions are included (Figure 6.12), the entropy dynamics become significantly more complex. The $(e_1 + \text{nuclei})$ partition now exhibits pronounced oscillations, indicating that correlations are no longer localized but instead redistributed across larger portions of the system. The (e_1, e_2) entropy also displays irregular, modulated oscillations, reflecting the presence of competing interaction scales and enhanced coupling between subsystems.

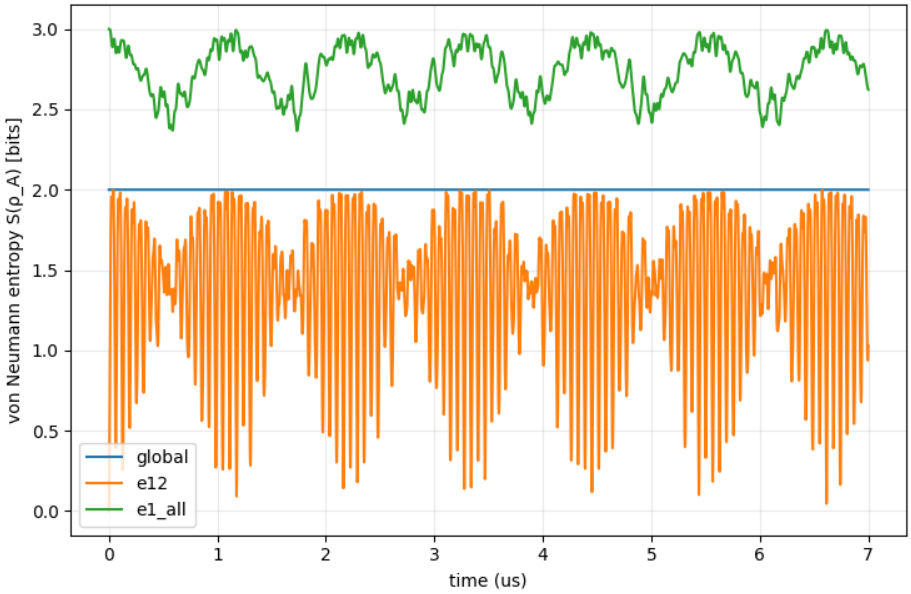


Figure 6.12 Subsystem von Neumann entropies with exchange and dipolar interactions. The dynamics become more complex, with significant redistribution of correlations across subsystems, indicating enhanced non-local entanglement.

This qualitative change demonstrates that exchange and dipolar couplings enable the electrons to act as a correlated pair rather than as independent subsystems. As a result, the nuclear spins effectively form a shared, structured environment, leading to non-trivial entanglement flow throughout the system.

6.3.4 Impact on singlet population dynamics.

The effect of these altered entanglement pathways is directly visible in the singlet population dynamics, shown in Figure 6.13. While both cases exhibit oscillatory behavior, the inclusion of exchange and dipolar interactions leads to more coherent and higher-amplitude oscillations, with the singlet population spending more time at larger values.

This behavior reflects the modified energy level structure introduced by the exchange and dipolar terms, which alter the singlet–triplet mixing frequencies and enable more efficient population transfer between spin states.

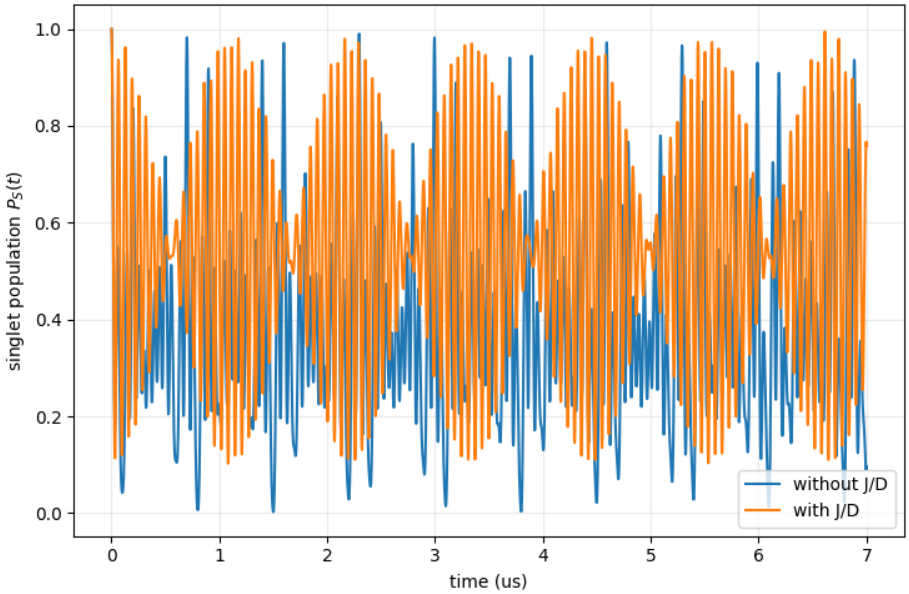


Figure 6.13 Singlet population dynamics with and without exchange/dipolar interactions. The inclusion of J and D leads to more structured and higher-amplitude oscillations.

6.3.5 Running and final singlet yield.

To quantify these differences, we examine the running time-averaged singlet yield,

$$\overline{P}_S(t) = \frac{1}{t}\int_0^t P_S(t')\,dt', \tag{6.2}$$

shown in Figure 6.14. The presence of exchange and dipolar interactions leads to a clear upward shift in the long-time average, with the system converging to a significantly higher steady value compared to the case without these couplings.

This difference is summarized in Figure 6.15, where the final time-averaged singlet yields are compared. The increase in yield is substantial, demonstrating that exchange and dipolar interactions are not merely perturbative corrections but instead play a dominant role in determining the observable output of the radical pair.

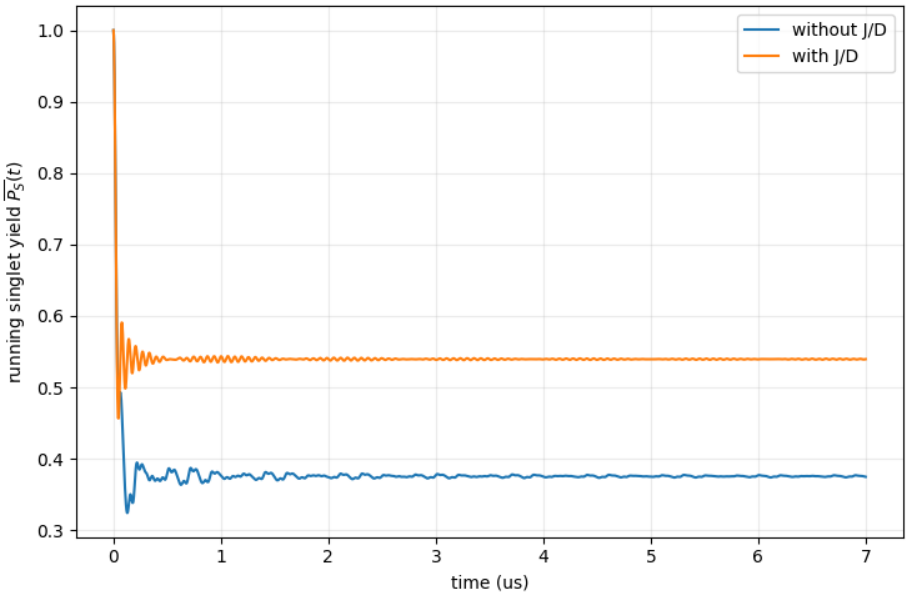


Figure 6.14 Running time-averaged singlet yield with and without exchange/dipolar interactions. The inclusion of J and D leads to a higher steady-state value.

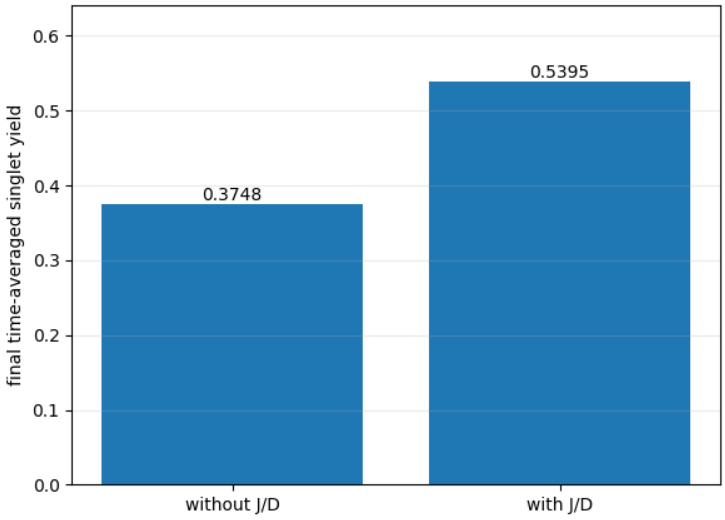


Figure 6.15 Final time-averaged singlet yield comparison. Exchange and dipolar interactions significantly enhance the singlet yield.

6.3.6 Exchange and Dipolar Robustness

Table 6.3 summarizes the distance robustness sweep performed in the simplified mixed-state model while preserving the cancellation condition $D = -3J$ at each separation. This test was designed to determine whether the interaction-driven singlet and subsystem-entropy phenomenology survives moderate

geometric perturbations when the absolute exchange–dipolar scale is changed but the cancellation ratio is held fixed. The results show that the qualitative behavior is robust across $r = 1.8$–$2.2\,\mathrm{nm}$. In particular, the final time-averaged singlet yield remains relatively stable, changing only from 0.5389 at 1.8 nm to 0.5237 at 2.2 nm, with a difference of just 1.39×10^{-2} relative to the $r = 2.0\,\mathrm{nm}$ baseline. Likewise, the singlet FFT remains highly stable across the sweep, with spectral RMS differences below 10^{-2} in both perturbed cases. This indicates that preserving the cancellation condition also preserves the dominant frequency structure of the singlet response, even as the absolute interaction strengths vary with distance.

Table 6.3 Distance robustness sweep with the cancellation condition preserved in the $D = -3J$ regime, using the simplified mixed-state entropy model with two nuclei coupled to electron 1. Simulations were performed at $B = 47\,\mu\mathrm{T}$ over a $5\,\mu\mathrm{s}$ window for $r = 1.8$, 2.0, and 2.2 nm. Shown are the final time-averaged singlet yield, the maximum subsystem entropies for the two partitions $A = (e_1e_2)|(\text{nuclei})$ and $B = (e_1 + \text{nuclei})|(e_2)$, and the RMS differences relative to the baseline $r = 2.0\,\mathrm{nm}$ case.

r (nm)	$\bar{\Phi}_S(T)$	$\max S_A$	$\max S_B$	Trace RMS	FFT RMS	S_A RMS	S_B RMS
1.8	0.5389	1.9933	3.0000	0.3589	0.0054	0.1446	0.1674
2.0	0.5376	1.9948	3.0000	0.0000	0.0000	0.0000	0.0000
2.2	0.5237	1.9830	3.0000	0.3238	0.0091	0.1329	0.1270

The subsystem entropies show a similar pattern of qualitative robustness with quantitative sensitivity. For the partition $A = (e_1e_2)|(\text{nuclei})$, the maximum entropy remains close to 2 bits for all three distances, while for the partition $B = (e_1 + \text{nuclei})|(e_2)$ the maximum entropy remains essentially fixed at 3 bits throughout the sweep. Thus, the overall mixed-state entropy scale and the existence of large oscillatory subsystem-entropy dynamics are preserved under moderate changes in radical-pair separation. At the same time, the entropy-trace RMS differences show that the detailed temporal organization of the subsystem correlations is still distance-sensitive: relative to the baseline case, the A-partition entropy differs by 0.1446 at 1.8 nm and 0.1329 at 2.2 nm, while the corresponding differences for partition B are 0.1674 and 0.1270. The singlet trace exhibits a comparable sensitivity, with RMS differences of 0.3589 and 0.3238 relative to the baseline. Taken together, these results show that preserving the cancellation condition preserves the qualitative singlet and entropy phenomenology, but not the detailed time-domain organization. Moderate geometric perturbations therefore do not destroy the behavior, yet the absolute interaction scale set by r remains a physically relevant control parameter.

Table 6.4 summarizes the robustness of the singlet and subsystem-entropy dynamics to departures from the exact cancellation condition at fixed geometry. In this test, the radical-pair separation was held at $r = 2.0\,\mathrm{nm}$ so that the dipolar interaction D remained fixed, while the exchange interaction J was varied to produce the ratios $D/J = -3.6, -3.3, -3.0, -2.7$, and -2.4. The exact-cancellation point $D/J = -3.0$ was used as the baseline. The results show that the qualitative singlet and entropy phenomenology survives

Table 6.4 Cancellation-ratio robustness sweep at fixed geometry, with $r = 2.0\,\mathrm{nm}$ and $B = 47\,\mu\mathrm{T}$. The dipolar interaction D was fixed by the chosen separation and the exchange interaction J was adjusted so that $D/J = -3.6, -3.3, -3.0, -2.7, -2.4$. Shown are the final time-averaged singlet yield, the maximum subsystem entropies for the two partitions $A = (e_1 e_2)|(\text{nuclei})$ and $B = (e_1 + \text{nuclei})|(e_2)$, and the RMS differences relative to the exact-cancellation baseline $D/J = -3.0$.

D/J	$\bar{\Phi}_S(T)$	$\max S_A$	$\max S_B$	Trace RMS	FFT RMS	S_A RMS	S_B RMS
-3.6	0.5472	1.9970	3.0000	0.3149	0.0370	0.6581	0.1756
-3.3	0.5432	1.9985	3.0000	0.2861	0.0211	0.6013	0.1241
-3.0	0.5395	1.9952	3.0000	0.0000	0.0000	0.0000	0.0000
-2.7	0.5346	1.9938	3.0000	0.3194	0.0242	0.6514	0.1409
-2.4	0.5289	1.9951	3.0000	0.3111	0.0464	0.6255	0.2270

across the entire sweep. In particular, all ratios retain strongly oscillatory singlet dynamics, structured Fourier spectra, and large-amplitude subsystem-entropy oscillations. This indicates that the behavior identified at the exact-cancellation point is not confined to a single finely tuned ratio.

At the same time, the table shows that the detailed time-domain organization is clearly sensitive to the cancellation detuning. Relative to the $D/J = -3.0$ baseline, the singlet-trace RMS differences remain large throughout the sweep, ranging from 0.2861 to 0.3194, while the entropy-trace RMS differences for partition $A = (e_1 e_2)|(\text{nuclei})$ are even larger, ranging from 0.6013 to 0.6581. The corresponding differences for partition $B = (e_1 + \text{nuclei})|(e_2)$ are smaller, but still significant, lying between 0.1241 and 0.2270. Thus, the cancellation point is not unique in producing the broad dynamical features, but it does act as an organizing point for the detailed timing and structure of both the singlet and subsystem-correlation dynamics.

By contrast, the time-averaged singlet yield varies only weakly across the sweep, decreasing smoothly from 0.5472 at $D/J = -3.6$ to 0.5289 at $D/J = -2.4$. The total variation in the final yield is therefore modest compared with the much larger changes seen in the singlet traces and entropy trajectories. The singlet FFT also remains relatively robust near the cancellation point, with the smallest spectral deviations occurring at $D/J = -3.3$ and -2.7, and larger deviations appearing only at the outer ratios. Finally, the maximum subsystem entropies remain essentially unchanged across the full sweep: $\max S_A$ stays close to 2 bits and $\max S_B$ remains fixed at 3 bits. This shows that moving away from exact cancellation does not destroy the system's ability to generate large subsystem mixing, but instead mainly reorganizes when and how that mixing appears in time. Taken together, these results show that the $D = -3J$ point is not an isolated anecdotal case, while also confirming that exact cancellation remains physically meaningful as a reference point for the detailed interaction-driven dynamics.

6.3.7 Physical interpretation.

The observed increase in singlet yield can be understood as a direct consequence of enhanced electron–electron correlations. In the absence of exchange and dipolar couplings, each electron primarily interacts with its local nuclear environment, limiting the extent of coherent singlet–triplet mixing. When these couplings are introduced, the electrons form a strongly interacting subsystem, enabling more efficient redistribution of population across spin states.

From an open-system perspective, this also implies that the nuclear environment acts as a more strongly coupled, non-Markovian bath when electron–electron interactions are present. The resulting feedback between subsystems allows for richer dynamical behavior and more efficient exploration of the system's Hilbert space.

These results highlight the importance of including exchange and dipolar interactions in realistic models of radical pair dynamics. Neglecting these terms can lead to qualitatively incorrect predictions, not only in the entanglement structure but also in experimentally relevant observables such as the singlet yield.

6.3.8 Zeeman Dependence of the Entanglement Envelope

The slow envelope modulation observed in the entropy dynamics suggests the presence of an additional dynamical frequency scale beyond the hyperfine interaction frequencies. At first glance, the envelope seemed to have a frequency mathcing that of the Zeeman interaction. To investigate the origin of this modulation, simulations were performed across a range of Zeeman interaction strengths. The oscillation frequency of the entropy envelope was extracted for each value of the Zeeman splitting and compared to the corresponding Zeeman frequency.

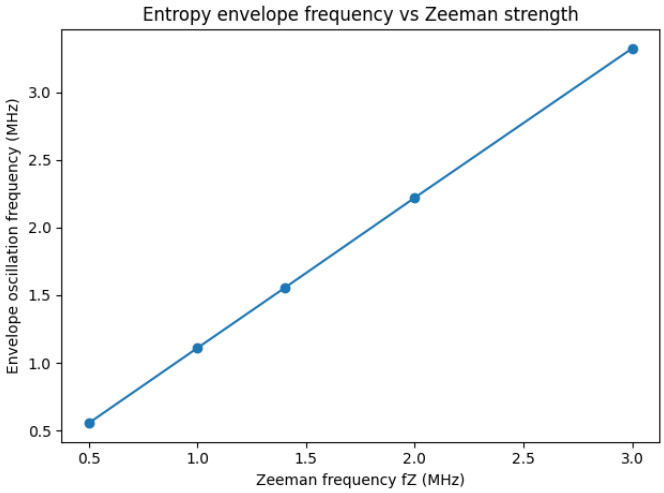


Figure 6.16 Entropy-envelope frequency versus Zeeman frequency. The slow beating frequency tracks the Zeeman scale approximately linearly over the field range examined.

The results in Figure 6.16 reveal an approximately linear relationship between the entropy envelope frequency and the Zeeman frequency over the range of field strengths examined. Near Earth-strength magnetic fields, the envelope frequency closely tracks the Zeeman splitting of the electron spins. This behavior indicates that the external magnetic field controls the large-scale flow of entanglement within the system, while the hyperfine interactions generate the faster oscillatory structure. Physically, the Zeeman interaction introduces an additional energy scale that competes with the hyperfine couplings. The resulting interference between these interactions produces the slow beating pattern observed in the subsystem entropies.

The entanglement dynamics observed in these simulations demonstrate that the nuclear environment plays a far more active role than is typically assumed in semiclassical radical pair models. Rather than acting as a static background field, the nuclei form a structured quantum reservoir that can: store quantum information, exchange information with the electrons, and generate coherent entanglement revivals on the order of microseconds. This behavior resembles the memory effects commonly associated with non-Markovian environments. However, in the radical pair system these effects arise naturally from the finite, discrete nuclear environment rather than from an external bath. These results therefore suggest a possible pathway toward Zeno-like stabilization mechanisms mediated by nuclear dynamics. If certain nuclear configurations or interactions effectively slow the flow of quantum information out of the electronic subsystem, they could prolong coherence or stabilize particular spin subspaces.

6.4 Controlled Nuclear Spin Operations and Spectral Restructuring

Building on the observation that the nuclear environment acts as a structured, information-storing reservoir for the radical pair system, we now investigate whether this environment can be actively manipulated to steer the spin dynamics. In particular, we examine the effect of localized unitary operations applied to individual nuclear spins during the evolution of the system. The motivation for this study is twofold. First, if the nuclear environment is capable of storing and returning quantum information—as demonstrated in Section 6.3—then targeted perturbations to specific nuclear degrees of freedom may allow controlled modification of the subsequent dynamics. Second, identifying which components of the nuclear environment exert the greatest influence on the radical pair provides insight into which physical mechanisms may be most relevant for controlling spin-dependent chemical outcomes. To this end, we introduce a controlled nuclear spin rotation in the form of a π-pulse, or 'spin-flip', applied to a selected nucleus at a fixed time during the evolution. The system is evolved unitarily under the full Hamiltonian described in Chapter 2, and at a fixed time ,during unitary evolution, of the form

$$U_{\text{flip}} = e^{-i\pi I_\alpha}$$

is applied to a single nuclear spin, where I_α is the spin operator along a chosen axis. The system is then evolved further under the same Hamiltonian. This protocol provides a controlled perturbation that allows us to probe the sensitivity of the radical pair dynamics to specific components of the nuclear environment. Simulations were performed using the six-nucleus model introduced in Section 6.2, in which each electron is coupled to three nuclei with heterogeneous hyperfine strengths: one strongly coupled nucleus, one moderately coupled nucleus, and one weakly coupled nucleus with coupling strength comparable to the Zeeman interaction. The flip operation was applied separately to two representative nuclei: (i) the strongly coupled nucleus and (ii) the weakly coupled nucleus with hyperfine strength on the order of the Zeeman splitting.

6.4.1 Strongly Coupled Nuclear Spin Flip

The effect of a π-pulse applied to the strongly coupled nucleus is shown in Figure 6.17. Following the pulse, a clear redistribution of population among the electronic spin subspaces is observed. The system transitions from oscillations about a pre-pulse steady configuration to a distinct post-pulse population balance, with a reduction in singlet population and a corresponding increase in triplet occupation. This shows that the control operation has redirected the system into a different region of accessible spin-states.

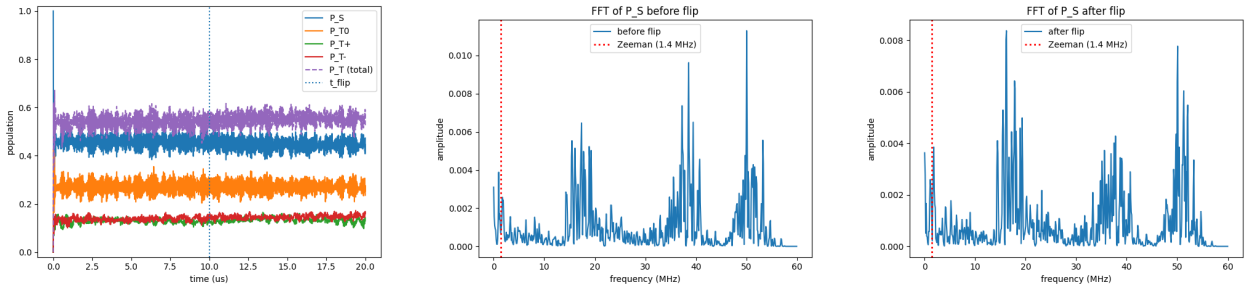


Figure 6.18 Effect of a π-pulse applied to a weakly coupled nucleus with hyperfine strength near the Zeeman scale. (a) Singlet and triplet populations before and after the pulse, showing minimal change in population dynamics. (b) Fourier spectrum of the singlet population before the pulse. (c) Fourier spectrum after the pulse. While the Zeeman-frequency peak remains stable, significant redistribution of higher-frequency spectral components is observed, indicating selective restructuring of the dynamical spectrum.

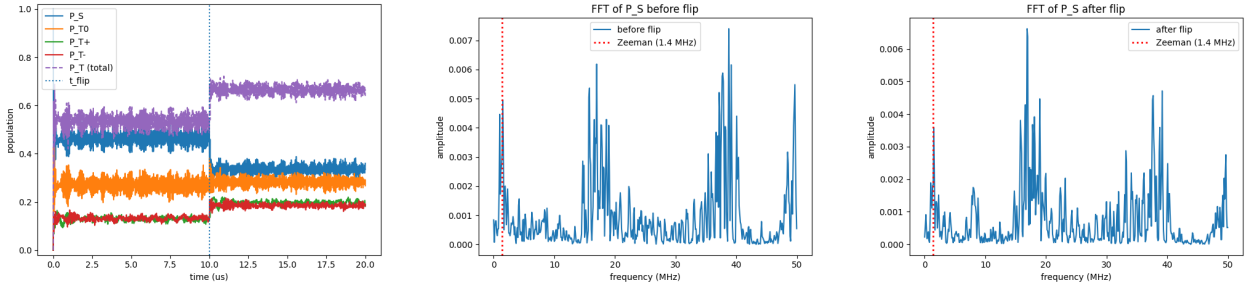


Figure 6.17 Effect of a π-pulse applied to a strongly coupled nucleus. (a) Singlet and triplet populations before and after the pulse applied at t_{flip}, showing clear redistribution of population among spin subspaces. (b) Fourier spectrum of the singlet population before the pulse. (c) Fourier spectrum after the pulse. The control operation primarily modifies higher-frequency components associated with strong hyperfine interactions, while preserving the low-frequency Zeeman-scale structure.

This behavior arises from the strong hyperfine coupling between the selected nucleus and the electron spin. The applied rotation modifies the joint electron–nuclear state, thereby altering the subsequent coherent evolution of the electronic subsystem. The corresponding Fourier spectra further clarify the nature of this control. While the low-frequency Zeeman-scale component remains present, the most significant changes occur in the higher-frequency structure of the spectrum. These higher-frequency components are associated with strong hyperfine interactions, indicating that the pulse predominantly perturbs the fast electron–nuclear dynamics. Together, these results demonstrate that strongly coupled nuclei act as population control channels, enabling substantial redistribution of electronic spin populations through localized nuclear operations.

6.4.2 Weakly Coupled Nuclear Spin Flip

The effect of applying the same π-pulse to a weakly coupled nucleus with hyperfine strength comparable to the Zeeman splitting is shown in Figure 6.18. In contrast to the strongly coupled case, the time-domain singlet and triplet populations exhibit minimal qualitative change following the pulse. The overall population balance between singlet and triplet subspaces remains largely unchanged, and the oscillatory structure of the dynamics is preserved. From a population-based perspective alone, the weak nucleus appears to exert only a minor influence on the system.

However, the frequency-domain analysis reveals a qualitatively different effect. The Fourier spectra of the singlet population before and after the pulse display a clear restructuring of spectral weight. In particular, the spectral feature associated with the Zeeman frequency (~ 1.4 MHz) remains largely unchanged, while substantial redistribution occurs among higher-frequency components. This indicates that the weakly coupled nucleus does not strongly affect population dynamics, but instead modifies how different dynamical frequencies contribute to the overall evolution.

These results reveal a clear distinction between the roles played by different components of the nuclear environment. Strongly coupled nuclei exert a dominant influence on the electronic spin populations, enabling direct redistribution of singlet and triplet occupation through local operations. In contrast, weak nuclei with hyperfine strengths near the Zeeman scale primarily influence the spectral structure of the dynamics, redistributing frequency components without inducing large population changes.

6.5 Nuclear Decay Dynamics

Having established that the nuclear environment acts as a structured, information-bearing subsystem capable of storing and redistributing quantum correlations (Sections 6.3–6.4), we now investigate a more extreme, and hypothetical, form of nuclear perturbation: a localized, non-unitary decay event.

Unlike the unitary π-pulse operations considered previously, nuclear decay introduces an irreversible quantum channel acting on the nuclear spin. As developed in Chapter 5, this process can be understood as generating entanglement between the remaining proton and emitted leptonic degrees of freedom, which are subsequently traced out. The net effect is a depolarizing transformation on the nuclear spin whose strength is controlled by the parameter q, representing the weight of the triplet-coupled decay channel.

The simulations presented here follow the same six-nucleus radical pair model introduced in Section 6.2. The system is initialized in the electronic singlet state with a maximally mixed nuclear environment, evolved under the Hamiltonian described in Chapter 2, subjected to a single nuclear decay event at $t_{\mathrm{decay}} = 2\ \mu$s, and then evolved further under the same Hamiltonian. To isolate the role of decay-induced entanglement, three representative values of q are considered: $q = 0.25$ (weak entanglement), $q = 0.50$

(moderate entanglement), and $q = 1.00$ (maximal entanglement within the model).

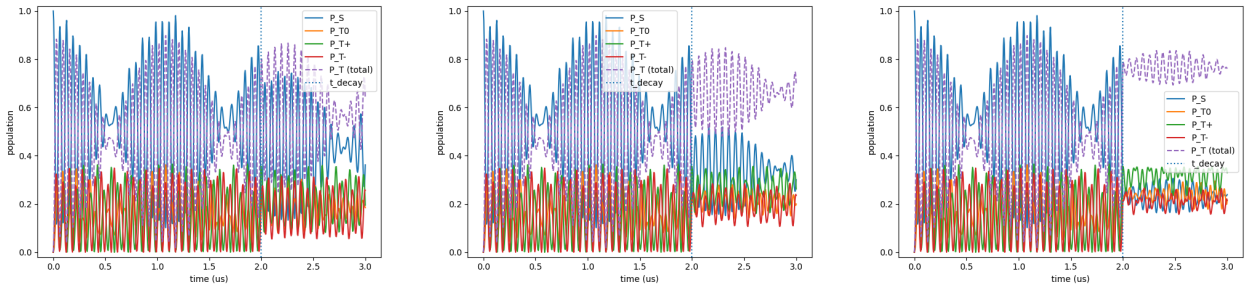


Figure 6.19 Singlet and triplet population dynamics before and after a single nuclear decay event at $t = 2$ μs, for varying levels of decay-induced entanglement: (a) $q = 0.25$, (b) $q = 0.50$, and (c) $q = 1.00$. The vertical dashed line indicates the time of the decay event. Populations shown include P_S, P_{T0}, P_{T+}, P_{T-}, and total triplet population.

Figure 6.19 summarizes the nuclear decay results across the three entanglement regimes. For small values of q, corresponding to weak entanglement between the proton and emitted leptons, the decay event induces only a modest perturbation to the system. Immediately following the decay, a slight redistribution of singlet and triplet populations is observed, but the overall oscillatory structure remains largely intact. This behavior is consistent with the interpretation of the decay as a weak depolarizing channel, in which the nuclear spin retains significant memory of its pre-decay state.

At intermediate values of q, the decay event produces a qualitatively stronger effect on the system dynamics. The amplitude of singlet oscillations is reduced, triplet populations redistribute toward a new dynamical balance, and the post-decay oscillations exhibit modified frequency content and reduced coherence. This regime reflects partial loss of information in the nuclear spin due to entanglement with the traced-out leptonic degrees of freedom.

In the limit of maximal entanglement within the model, the decay produces a dramatic restructuring of the spin dynamics. The decay event effectively resets the nuclear spin to a highly mixed state, resulting in strong suppression of coherent singlet–triplet oscillations, significant redistribution of population toward triplet subspaces, and a marked change in the long-time dynamical behavior of the system. However, even in this limit, residual oscillatory structure persists, indicating that the remaining nuclei continue to participate in the dynamics. The decay therefore acts as a localized disruption rather than a global loss of coherence.

A central open question in this model is the physical magnitude of the entanglement between the remaining proton and the emitted leptons. While the parameter q provides a convenient way to interpolate between different regimes, its realistic value for nuclear decay processes remains uncertain. Direct

measurement of this entanglement is experimentally challenging, particularly due to the difficulty of detecting and performing joint measurements on emitted antineutrinos. As a result, the degree of entanglement is treated here as a free parameter. The results suggest a potential indirect probe: the singlet and triplet population dynamics of the radical pair may serve as a sensitive diagnostic of the entanglement generated during decay. In particular, increasing q leads to systematic suppression and restructuring of coherent oscillations, distinct dynamical signatures emerge across different entanglement regimes, and these signatures persist over microsecond timescales relevant to radical pair reactions. This raises the possibility that spin-selective observables within the radical pair system could, in principle, encode information about otherwise inaccessible quantum correlations in the decay products.

6.6 Statistical Error Analysis

Table 6.5 Numerical convergence under timestep refinement for a fixed $5\,\mu$s simulation window.

Model	Comp.	Trace RMS	Yield (ref)	$\lvert\Delta$Yield$\rvert$	FFT RMS
Full QM	A vs B	6.14×10^{-4}	0.46358327	2.47×10^{-7}	8.76×10^{-4}
Full QM	B vs C	1.91×10^{-3}	0.46358302	1.06×10^{-6}	2.85×10^{-3}
Overhauser	A vs B	3.82×10^{-4}	0.42696388	2.06×10^{-7}	9.42×10^{-4}
Overhauser	B vs C	9.69×10^{-5}	0.42696368	5.22×10^{-8}	5.03×10^{-4}

Timestep refinement tests were performed for both the full quantum and Overhauser ensemble models over a fixed $5\,\mu$s simulation window using $dt = 0.002$, 0.001, and $0.0005\,\mu$s. In both cases, the final time-averaged singlet yield was numerically stable, changing by no more than 10^{-6} across the refinement ladder. The Overhauser model additionally showed decreasing singlet-trace and FFT spectral RMS differences under refinement, confirming numerical convergence. The full quantum model displayed similarly stable singlet traces and spectra by visual inspection, although direct peak-by-peak FFT matching was found to be unreliable due to rank reordering among nearby spectral features. Overall, the results support the use of $dt = 0.002\,\mu$s as a numerically adequate baseline timestep for the $5\,\mu$s simulations. Floating-point roundoff was not found to be a dominant source of uncertainty; numerical robustness was instead assessed through convergence checks with respect to solver settings, temporal resolution, and sampling.

Table 6.6 Ensemble convergence of the Overhauser simulations for a fixed $5\,\mu$s runtime with $dt = 0.002\,\mu$s.

N	$\bar{\Phi}_S(T)$	Std. err.	Trace RMS to prev.	$\lvert\Delta\bar{\Phi}_S(T)\rvert$ to prev.	FFT RMS to prev.
50	0.418712	0.020407	–	–	–
100	0.420707	0.013006	0.024277	0.001995	0.167295
200	0.418047	0.009476	0.017638	0.002659	0.187111
400	0.426964	0.007179	0.015371	0.008916	0.182473
800	0.427668	0.004845	0.008755	0.000704	0.163103

The Overhauser ensemble convergence was assessed by increasing the number of sampled realizations from $N = 50$ to $N = 800$ while holding all other parameters fixed. The ensemble-averaged singlet traces became progressively smoother with increasing N, and the RMS difference between consecutive traces decreased overall, reaching 8.76×10^{-3} for 400 vs 800. The final time-averaged singlet yield also stabilized at large N, with the difference between $N = 400$ and $N = 800$ reduced to 7.04×10^{-4}. At the same time, the standard error of the final yield estimate decreased monotonically from 2.04×10^{-2} at $N = 50$ to 4.85×10^{-3} at $N = 800$. These results indicate that the Overhauser ensemble average is largely converged by $N \approx 400$, supporting the use of $N = 400$ as a reasonable baseline for the main Overhauser simulations.

CHAPTER 7
DISCUSSION

7.1 Overview and Physical Picture

The results presented in Chapter 6 establish a consistent physical picture of the radical pair system when treated within a fully quantum mechanical framework. Across all investigations—thermalization, semiclassical comparison, entanglement dynamics, controlled nuclear operations, and nuclear decay—the same underlying structure emerges: the nuclear environment is not a passive background, but an active quantum subsystem that participates directly in the system dynamics.

This conclusion is not inferred from a single observable, but rather from the convergence of several independent diagnostics. The failure of the Overhauser approximation, the persistence of subsystem entropy oscillations, the appearance of entanglement revivals, and the sensitivity of the system to localized nuclear operations all point toward the same interpretation. The radical pair must be understood as a coupled electron–nuclear system in which quantum information is continuously exchanged between subsystems.

From this perspective, the nuclear spins act as a finite reservoir with internal structure. Unlike an idealized bath, this reservoir does not irreversibly absorb information. Instead, it stores quantum correlations and returns them to the electronic subsystem on characteristic timescales. This behavior underlies many of the phenomena observed in this work, including the structured oscillations in subsystem entropy and the emergence of slow envelope modulations in the entanglement dynamics.

7.2 Open System Treatment

Although the open system treatment based on spontaneous emission with best-estimate biological jump rates yielded thermalization times on the order of milliseconds, we do not expect radical-pair coherence times of that magnitude. Having neglected spin-selective recombination dynamics, structured or non-Markovian environmental modes, stronger system–bath correlations beyond the Born–Markov regime, non-spontaneous-emission decoherence channels such as pure dephasing, it is clear that more work and consideration would need to go into an open-system model that would accurately capture coherence times of radical pairs involved in magnetoreception. This work hopes to lay some of the foundation for such open-system approaches in the radical pair model and the results from these simulations show that spontaneous emission is unlikely to play a significant role in the dynamics of the system on the microsecond coherence times expected of the radical pair.

7.3 Breakdown of Semiclassical Descriptions

One of the central motivations of this work was to assess the validity of semiclassical approaches to radical-pair dynamics. The comparison with the Overhauser-field approximation provides a clear answer: while semiclassical models can reproduce certain averaged features, they fail to capture the underlying dynamical structure of the system. In particular, replacing the nuclear spins with a classical stochastic field removes several defining features of the full quantum model. The coherent oscillatory structure generated by discrete hyperfine couplings is lost, the many-body spin network no longer supports the same multi-frequency dynamics, and the entanglement and information exchange between the electronic and nuclear subsystems disappear from the description. These missing features are not minor corrections. They qualitatively alter the behavior of the system and, as shown in Chapter 6, remain important even in regimes where ensemble-averaged observables appear to agree reasonably well.

This breakdown is most pronounced when discrete nuclear couplings compete with other interaction scales, such as the Zeeman, exchange, and dipolar interactions. In these regimes, the system operates in a genuinely mesoscopic domain, where neither purely classical nor fully thermodynamic descriptions are adequate.

An additional subtlety in the breakdown of semiclassical descriptions arises when considering time-averaged observables. As shown in Section 6.2, the Overhauser approximation can, in some cases, reproduce singlet populations and their time-averaged values with reasonable accuracy, particularly under ensemble averaging of maximally mixed nuclear states. At the level of coarse-grained observables, this may suggest that the semiclassical model provides an adequate description of the system.

However, this apparent agreement masks a more significant failure in the underlying dynamics. The full quantum model exhibits a rich, multi-frequency structure arising from coherent electron–nuclear interactions, including components near the Zeeman frequency that are essential for magnetic-field sensitivity. In contrast, the Overhauser approximation suppresses or distorts this spectral structure, replacing it with either broadened or incoherent frequency contributions.

This distinction is critical in the context of magnetoreception, where sensitivity to weak external magnetic fields is mediated by well-defined low-frequency spin dynamics. While time-averaged singlet populations may appear similar, the loss of structured frequency content in the semiclassical model implies a reduced ability to accurately capture the mechanisms responsible for directional sensitivity. Consequently, agreement at the level of averaged observables does not imply that the semiclassical description preserves the functional behavior required for biologically relevant magnetoreception.

7.4 Nuclear Spins as a Non-Markovian Environment

A key result of this work is that non-Markovian behavior arises intrinsically from the finite nuclear environment, even in the absence of an external bath. This is most clearly seen in the entanglement dynamics, where subsystem entropy exhibits oscillatory behavior with clear revivals.

These revivals indicate that quantum information is not irreversibly lost from the electronic subsystem. Instead, it flows into the nuclear degrees of freedom and is later returned. This bidirectional flow is the defining feature of non-Markovian dynamics.

An important connection between the internal quantum dynamics and experimentally relevant observables is provided by the time-averaged singlet yield. While instantaneous singlet populations capture short-time coherent oscillations, the running singlet yield integrates these dynamics over time, effectively encoding the cumulative impact of singlet–triplet interconversion processes.

The results of Section 6.3 show that the inclusion of exchange and dipolar interactions leads to a substantial increase in the long-time singlet yield. This demonstrates that modifications to the underlying entanglement structure are not merely abstract features of the quantum state, but have direct and measurable consequences. In particular, the enhanced electron–electron correlations introduced by these interactions enable more efficient redistribution of population within the singlet–triplet manifold, resulting in a higher steady-state singlet yield.

From this perspective, the singlet yield can be viewed as a coarse-grained probe of the system's entanglement dynamics. The sensitivity of the yield to interaction terms such as exchange and dipolar couplings highlights the importance of accurately capturing these effects in any realistic model of radical pair systems. It also suggests that observable chemical outputs may provide indirect access to the underlying quantum information dynamics, including correlation flow and non-Markovian memory effects.

Importantly, this behavior emerges under purely unitary evolution. No explicit non-Markovian master equation is required. The memory effects arise from the discrete and finite nature of the nuclear spin system itself.

The inclusion of exchange and dipolar interactions plays a crucial role in revealing this structure. As shown in Chapter 6, these interactions introduce an additional energy scale that modulates the flow of entanglement, producing slow envelope oscillations. The observed linear relationship between the envelope frequency and the Zeeman splitting indicates that the external magnetic field controls the large-scale structure of information flow, while hyperfine interactions generate the fast dynamics.

This interplay between interaction scales highlights a key point: non-Markovianity in radical pair systems is not an externally imposed feature, but an intrinsic consequence of the system's internal

structure.

7.5 Timescales and the Limited Role of External Baths

The open-system simulations further clarify the role of environmental coupling. While the GKSL dynamics drive the system toward a steady state, the associated timescales are orders of magnitude longer than those governing coherent spin dynamics.

As shown in Chapter 6, coherent oscillations occur on microsecond timescales set by hyperfine and Zeeman interactions, while relaxation toward equilibrium occurs on millisecond timescales. This separation implies that, over the lifetime of the radical pair, the system retains significant memory of its initial state.

This result places strong constraints on the relevance of Markovian environmental models in magnetoreception. While such models are useful for describing long-time behavior, they do not dominate the dynamics on biologically relevant timescales. Instead, the primary source of dynamical structure is the internal nuclear environment.

This observation supports a shift in modeling strategy: rather than focusing on external baths as the primary source of decoherence, it is more appropriate to treat the nuclear spins themselves as the dominant environment.

7.6 Control Through the Nuclear Environment

Beyond passive dynamics, the results demonstrate that the nuclear environment can serve as a mechanism for control. Localized nuclear spin operations produce significant and highly structured changes in both population dynamics and spectral content.

A clear distinction emerges between strongly and weakly coupled nuclei. Operations on strongly coupled nuclei primarily affect population distributions, while operations on weakly coupled nuclei influence the frequency content of the dynamics. This separation suggests that different components of the nuclear environment can act as distinct control channels.

While the specific control protocols considered here are theoretical, they illustrate a broader principle: internal degrees of freedom within the radical pair system can be used to steer its dynamics. This raises the possibility that biological systems could exploit similar mechanisms, although identifying physically realizable implementations remains an open challenge.

7.7 Nuclear Decay as a Quantum Channel

The introduction of a quantum-information-based model of nuclear decay extends the role of the nuclear environment beyond coherent dynamics. In this framework, decay acts as a localized quantum channel whose strength is determined by the entanglement generated with external degrees of freedom.

The simulations show that even a single decay event can significantly restructure the system dynamics. As the entanglement parameter increases, coherent oscillations are suppressed and population distributions are altered. At the same time, the persistence of residual structure indicates that the remaining nuclear spins continue to support coherent dynamics.

This behavior demonstrates that processes external to the radical pair system can influence its dynamics indirectly through their effect on the nuclear environment. It also suggests that observable quantities, such as singlet and triplet populations, may encode information about otherwise inaccessible quantum correlations.

While the present model is intentionally simplified, it provides a proof-of-principle demonstration that the nuclear environment can mediate the impact of external quantum processes on radical pair dynamics.

7.8 Implications for Radical Pair Mechanisms

Taken together, these results support a revised view of radical pair systems. Rather than treating the nuclei as a source of noise, they should be regarded as an integral component of the quantum dynamics.

Several implications follow from this perspective.

First, quantum coherence and entanglement can persist over timescales relevant to radical pair reactions, supported by the structured dynamics of the nuclear environment. This challenges the assumption that biological environments necessarily suppress quantum effects on short timescales.

Second, the identification of the nuclear environment as an intrinsic non-Markovian subsystem highlights the importance of finite, structured environments in biological quantum systems. Models that neglect this structure risk missing essential dynamical features.

Third, the possibility of control through nuclear degrees of freedom suggests new avenues for both biological function and artificial design. Even if biological systems do not actively implement such control, the underlying mechanisms are present within the physics of the system.

Finally, the coupling between internal dynamics and external quantum processes, as illustrated by the decay model, points toward a broader view of radical pairs as components of larger quantum networks embedded within complex environments.

7.9 Limitations and Outlook

While the results of this work establish a clear qualitative picture, several limitations should be acknowledged.

The nuclear environments considered here are relatively small and simplified. Extending the model to larger spin networks with realistic hyperfine tensors remains an important direction for future work.

The open-system treatment is restricted to Markovian dynamics. Incorporating explicitly non-Markovian baths would allow direct comparison with the intrinsic memory effects identified in this work.

The nuclear decay model is highly idealized and does not include detailed nuclear physics. A more complete treatment would be required to assess its physical relevance.

Finally, experimental validation remains challenging. Direct access to electron–nuclear entanglement is limited, and indirect observables must be interpreted with care.

Despite these limitations, the framework developed here provides a foundation for studying radical pair systems as mesoscopic quantum systems governed by structured information flow.

CHAPTER 8
CONCLUSION AND OUTLOOK

This thesis set out to investigate the extent to which radical pair systems require a fully quantum mechanical description and to determine how quantum correlations, environmental structure, and nuclear dynamics influence their behavior. By constructing and analyzing a many-body spin model incorporating a full quantum Hamiltonian, and both Markovian and non-Markovian environmental effects, this work has established a coherent picture of radical pair dynamics as a problem of quantum information flow rather than purely chemical kinetics.

8.1 Conclusion

The results demonstrate that, within biologically relevant regimes, radical pair dynamics are overwhelmingly governed by coherent Hamiltonian interactions. Open-system effects modeled through GKSL dynamics were shown to act on timescales far longer than the microsecond lifetimes of radical pairs, rendering spontaneous-emission–type dissipation largely irrelevant for magnetoreception. Consequently, coherent dynamics driven by weak environmental coupling are unlikely to play a significant role in realistic systems. Instead, the dominant features of the dynamics arise from the internal structure of the spin system itself, particularly the interplay between hyperfine, Zeeman, exchange, and dipolar interactions .

A central conclusion of this work is that the nuclear environment cannot be accurately treated as a passive or classical background. Comparisons with the Overhauser field approximation revealed that while semiclassical models capture coarse features of the dynamics, they fail to reproduce the full spectral and temporal structure generated by discrete nuclear spins. In particular, explicitly quantum nuclear environments support multi-frequency dynamics, coherent oscillatory structure, and dynamical correlations that are absent in classical field descriptions. These findings establish that the nuclear environment must be treated as an active quantum subsystem whenever hyperfine interactions compete with other energy scales in the system.

Perhaps the most significant insight of this thesis is the identification of the nuclear environment as an intrinsic source of non-Markovian behavior. Even in the absence of an external bath, the finite nuclear spin system acts as a structured reservoir capable of storing and returning quantum information to the electronic subsystem. This manifests as entanglement exchange, revivals, and oscillatory information flow between electrons and nuclei. The observed correlation dynamics demonstrate that non-Markovian effects in radical pair systems need not arise from complex external environments, but can instead emerge naturally from the internal many-body structure of the system.

The inclusion of exchange and dipolar interactions further revealed that the qualitative nature of the dynamics is highly sensitive to interaction regimes. In particular, the cancellation conditions identified by Efimova and Hore introduce degeneracies that amplify the influence of weak magnetic fields, while simultaneously modifying entanglement dynamics and correlation transport. These results highlight the importance of considering the full Hamiltonian structure when analyzing radical pair systems, as simplified models may overlook regimes where competing interactions produce qualitatively different behavior.

Beyond passive dynamics, this work also demonstrated that the nuclear environment provides a viable pathway for controlling radical pair spin dynamics. Localized nuclear operations, such as spin flips, were shown to induce significant changes in both population dynamics and spectral structure. Strongly coupled nuclei primarily influence population redistribution, while weakly coupled nuclei near resonance with the Zeeman interaction can selectively modify frequency content. This suggests that different components of the nuclear environment may serve as distinct control channels, offering a route toward engineered manipulation of spin dynamics in both biological and synthetic systems.

The introduction of a quantum-information-theoretic model of nuclear decay further extended this framework by incorporating discrete, non-unitary processes into the spin dynamics. By modeling decay as an isometric map that generates entanglement with emitted particles and induces a depolarizing channel on the remaining nucleus, this work demonstrated how localized quantum events can propagate through the system and produce observable changes in electronic spin dynamics. This approach provides a novel conceptual bridge between particle decay processes and quantum information theory, illustrating how even simple phenomenological models can reveal nontrivial dynamical consequences .

Taken together, these results support a unified interpretation of radical pair systems as mesoscopic quantum networks in which information is continuously exchanged between electrons and nuclei. The dynamics are shaped not only by external fields and environmental noise, but also by the internal structure and connectivity of the spin system. In this regime, quantum correlations are not incidental but play a central role in determining system behavior.

In conclusion, this thesis demonstrates that radical pair systems occupy a regime where quantum mechanics is not merely a correction to classical behavior, but a fundamentally necessary framework for understanding their dynamics. The interplay between coherent interactions, structured environments, and quantum correlations places these systems at the boundary between quantum information science and biological physics, offering both conceptual insight and potential avenues for control and application.

8.2 Outlook

Looking forward, several directions emerge for future work. A key extension is the incorporation of structured, few-mode environments beyond the Markovian GKSL framework to more fully explore non-Markovian dynamics arising from external degrees of freedom. Additionally, connecting the quantum correlation dynamics identified in this work to experimentally measurable quantities, such as reaction yields or spectroscopic signatures, remains an important step toward experimental validation. While quantities such as entanglement provide deep insight into the internal structure of the dynamics, they are not directly measurable in biological systems. Bridging this gap will require systematically relating these quantities to singlet yields, reaction kinetics, or spectroscopic signatures. In particular, the sensitivity of frequency-domain features and time-averaged observables to nuclear structure suggests that experimental measurements may already encode indirect signatures of quantum correlations. Developing this mapping would provide a pathway toward experimentally validating the quantum-information-theoretic picture of radical pair dynamics. Furthermore, the possibility of engineering nuclear environments—either through molecular design or external control—to enhance or exploit quantum effects represents a promising avenue for both quantum biology and quantum technology.

Perhaps the most exciting next step for this modeling approach is to extend the same methods to other biological realizations of the RPM. Our model is based specifically on the $FAD^{\cdot -}$–$TrpH^{\cdot +}$ RPM on cryptochrome where specific molecule dimensions and interaction strengths determine the dynamics of the system. That being said, there are a range of molecular candidates for the RPM which would have different different geometries, distances, interaction strengths and environments that could take the RPM into separate regimes of interaction dominance and steer-ability. All of which could apply apply the same quantum-information-theoretic approach. Ideally, this approach could produce a systematic "regime map" for radical pair dynamics. The results of this thesis indicate that the qualitative behavior of the system depends sensitively on the relative magnitudes of hyperfine, Zeeman, exchange, and dipolar interactions. Mapping out these regimes—identifying where semiclassical approximations fail, where non-Markovian effects dominate, and where control through nuclear degrees of freedom becomes effective—would provide a unifying framework for comparing different molecular systems. Such a classification could also guide the design of artificial radical pair systems optimized for specific dynamical or sensing properties.

An additional avenue for future work is the extension of the control framework developed here beyond single, localized spin operations to fully time-dependent driving protocols. While the -pulse experiments demonstrate that individual nuclear spins can act as effective control channels, more sophisticated control schemes—such as periodic pulse sequences, continuous-wave driving, or phase-engineered pulse

trains—could enable selective manipulation of specific dynamical features. For example, regularly spaced pulses may be used to stabilize or suppress particular frequency components, while continuous driving fields could be tuned to induce level crossings or resonant mixing between spin states. Similarly, sequences of inverted or time-reversed pulses raise the possibility of observing echo-like phenomena, providing a direct probe of coherence and information backflow within the electron–nuclear system. Developing such control protocols would not only deepen understanding of the system's dynamical structure, but also establish a foundation for coherent manipulation of radical pair systems using techniques analogous to those in magnetic resonance and quantum control.

More broadly, this work suggests that radical pair systems occupy a unique position at the interface of quantum information science, chemistry, and biology. As experimental techniques continue to improve, enabling greater control and measurement of spin dynamics in complex molecular systems, the conceptual framework developed here may provide a foundation for interpreting these systems not merely as chemical reactions, but as structured quantum networks. Exploring this perspective further may reveal new principles governing the role of quantum mechanics in biological function and inspire new approaches to harnessing quantum effects in noisy, mesoscopic environments.

REFERENCES

[1] Pascual Jordan. *Die Physik des 20. Jahrhunderts*. Vieweg, 1932.

[2] Erwin Schrödinger. *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.

[3] Neill Lambert, Yueh-Nan Chen, Yuan-Chung Cheng, Che-Ming Li, Guang-Yin Chen, and Franco Nori. Quantum biology. *Nature Physics*, 9:10–18, 2013. doi: 10.1038/nphys2474.

[4] P. J. Hore and H. Mouritsen. The radical-pair mechanism of magnetoreception. *Annual Review of Biophysics*, 45:299–344, 2016. doi: 10.1146/annurev-biophys-032116-094545.

[5] Markus Arndt, Thomas Juffmann, and Vlatko Vedral. Quantum physics meets biology. *HFSP Journal*, 3:386–400, 2009. doi: 10.2976/1.3244985.

[6] Gregory D Scholes and Graham R Fleming. What is quantum biology? *Proceedings of the National Academy of Sciences*, 123(14):e2531134123, 2026.

[7] K. Schulten and P. G. Wolynes. Semiclassical description of electron spin motion in radicals including the effect of electron hopping. *J. Chem. Phys.*, 68:3292, 1978.

[8] Christopher T. Rodgers and P. J. Hore. Chemical magnetoreception in birds: The radical pair mechanism. *PNAS*, 106:353–360, 2009. doi: 10.1073/pnas.0711968106.

[9] Jianming Cai, Gian Giacomo Guerreschi, and Hans J. Briegel. Quantum control and entanglement in a chemical compass. *Physical Review Letters*, 104:220502, 2010. doi: 10.1103/PhysRevLett.104.220502.

[10] Erik M. et al. Gauger. Sustained quantum coherence and entanglement in the avian compass. *Physical Review Letters*, 106:040503, 2011. doi: 10.1103/PhysRevLett.106.040503.

[11] Farhan T Chowdhury, Matt CJ Denton, Daniel C Bonser, and Daniel R Kattnig. Quantum control of radical-pair dynamics beyond time-local optimization. *PRX Quantum*, 5(2):020303, 2024.

[12] Igor A. Solov'yov, David E. Chandler, and Klaus Schulten. Magnetic field effects in arabidopsis thaliana cryptochrome-1. *Biophysical Journal*, 107:1743–1756, 2014. doi: 10.1016/j.bpj.2014.07.040.

[13] Kenichi Maeda, Kevin B Henbest, Federico Cintolesi, Ilya Kuprov, Christopher T Rodgers, Paul A Liddell, Devens Gust, Christian R Timmel, and P J Hore. Molecular mechanism of light-induced signaling by cryptochrome. *Nature*, 453:387–390, 2008.

[14] Junjie Chen and Yun Chen. Controlling cell behavior with engineered magnetosensitive proteins. *Biophysical Journal*, 123(3):20a, 2024.

[15] Katherine M Xiang, Hana Lampson, Rebecca Frank Hayward, Andrew G York, Maria Ingaramo, and Adam E Cohen. Mechanism of giant magnetic field effect in fluorescence of mscarlet3, a red fluorescent protein. *bioRxiv*, pages 2025–02, 2025.

[16] Shiny Maity, Jackson Sheppard, Hannah Russell, Chung-Ta Han, Leah Epstein, Bruce A Johnson, Brad Price, Ruixian Han, Lokeswara Rao Potnuru, Jinlei Cui, et al. Hydraulic activation of the aslov2 photoreceptor. *bioRxiv*, 2025.

[17] Gabriel Abrahams, Ana Štuhec, Vincent Spreng, Robin Henry, Idris Kempf, Jessica James, Kirill Sechkar, Scott Stacey, Vicente Trelles-Fernandez, Lewis M Antill, et al. Quantum spin resonance in engineered magneto-sensitive fluorescent proteins enables multi-modal sensing in living cells. *bioRxiv*, pages 2024–11, 2024.

[18] Noboru Ikeya and Jonathan R Woodward. Cellular autofluorescence is magnetic field sensitive. *Proceedings of the National Academy of Sciences*, 118(3):e2018043118, 2021.

[19] Noboru Ikeya and Jonathan R Woodward. A fluorescence microscopy platform for time-resolved studies of spin-correlated radical pairs in biological systems. *Journal of the American Chemical Society*, 2026.

[20] Hiroshi Yukawa, Hidetoshi Kono, Hitoshi Ishiwata, Ryuji Igarashi, Yoichi Takakusagi, Shigeki Arai, Yu Hirano, Tetsuya Suhara, and Yoshinobu Baba. Quantum life science: biological nano quantum sensors, quantum technology-based hyperpolarized mri/nmr, quantum biology, and quantum biotechnology. *Chemical Society Reviews*, 54(7):3293–3322, 2025.

[21] Hermann Edlbauer, Junliang Wang, AM Saffat-Ee Huq, Ian Thorvaldson, Michael T Jones, Saiful Haque Misha, William J Pappas, Christian M Moehle, Yu-Ling Hsueh, Henric Bornemann, et al. An 11-qubit atom processor in silicon. *Nature*, 648(8094):569–575, 2025.

[22] Hamish G Hiscock, Susannah Worster, Daniel R Kattnig, Charlotte Steers, Ye Jin, David E Manolopoulos, Henrik Mouritsen, and Peter J Hore. The quantum needle of the avian magnetic compass. *Proceedings of the National Academy of Sciences*, 113(17):4634–4639, 2016.

[23] Christian R. Timmel, Ulrich Till, Brian Brocklehurst, Keith A. McLauchlan, and P. J. Hore. Effects of weak magnetic fields on free radical recombination reactions. *Molecular Physics*, 95:71–89, 1998.

[24] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.

[25] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, 2002.

[26] Olga Efimova and P. J. Hore. Role of exchange and dipolar interactions in the radical pair model. *Biophysical Journal*, 94:1565–1574, 2008. doi: 10.1529/biophysj.107.118588.

[27] Thorsten Ritz, Peter Thalau, John B Phillips, Roswitha Wiltschko, and Wolfgang Wiltschko. Resonance effects indicate a radical-pair mechanism for avian magnetic compass. *Nature*, 429:177–180, 2004.

[28] S. et al. Engels. Anthropogenic electromagnetic noise disrupts magnetic compass orientation in a migratory bird. *Nature*, 509:353–356, 2014. doi: 10.1038/nature13290.

[29] Ulrich E. Steiner and Thomas Ulrich. Magnetic field effects in chemical kinetics and related phenomena. *Chemical Reviews*, 89:51–147, 1989. doi: 10.1021/cr00091a003.

[30] Nikolai V. Prokof'ev and P. C. E. Stamp. Theory of the spin bath. *Reports on Progress in Physics*, 63: 669–726, 2000. doi: 10.1088/0034-4885/63/4/204.

[31] R. Haberkorn. Density matrix description of spin-selective radical pair reactions. *Mol. Phys.*, 32: 1491–1493, 1976.

[32] Jason Lau, Hannah J Hogben, Till Biskup, Daniel R Kattnig, PJ Hore, et al. Alternative radical pairs for cryptochrome-based magnetoreception. *Journal of The Royal Society Interface*, 11(95), 2014.

[33] Bo Leberecht, Siu Ying Wong, Baladev Satish, Sara Döge, Jakob Hindman, Lalitha Venkatraman, Shambhavi Apte, Katrin Haase, Isabelle Musielak, Glen Dautaj, Ilia A. Solov'yov, Michael Winklhofer, Henrik Mouritsen, and P. J. Hore. Upper bound for broadband radiofrequency field disruption of magnetic compass orientation in night-migratory songbirds. *Proceedings of the National Academy of Sciences*, 120(28):e2301153120, 2023. doi: 10.1073/pnas.2301153120.

[34] Betony Adams, Ilya Sinayskiy, and Francesco Petruccione. An open quantum system approach to the radical pair mechanism. *Scientific reports*, 8(1):15719, 2018.

[35] A. Abragam. *The Principles of Nuclear Magnetism*. Oxford University Press, 1961.

[36] Thorsten Ritz, Salih Adem, and Klaus Schulten. A model for photoreceptor-based magnetoreception in birds. *Biophysical Journal*, 78:707–718, 2000. doi: 10.1016/S0006-3495(00)76629-X.

[37] Hideo Hayashi. *Introduction to Dynamic Spin Chemistry*. World Scientific, 2004.

[38] J. H. Van Vleck. The dipolar broadening of magnetic resonance lines in crystals. *Phys. Rev.*, 74: 1168–1183, Nov 1948. doi: 10.1103/PhysRev.74.1168. URL https://link.aps.org/doi/10.1103/PhysRev.74.1168.

[39] Olga Efimova and PJ Hore. Evaluation of nuclear quadrupole interactions as a source of magnetic anisotropy in the radical pair model of the avian magnetic compass. *Molecular Physics*, 107(7): 665–671, 2009.

[40] John Von Neumann. *Mathematical foundations of quantum mechanics: New edition*. Princeton university press, 2018.

[41] Ugo Fano. Description of states in quantum mechanics by density matrix and operator techniques. *Reviews of modern physics*, 29(1):74, 1957.

[42] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan. Completely positive dynamical semigroups. *Journal of Mathematical Physics*, 17:821, 1976.

[43] G. Lindblad. On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48:119, 1976.

[44] A. Kossakowski. On quantum statistical mechanics of non-hamiltonian systems. *Reports on Mathematical Physics*, 3(4):247–274, 1972. doi: 10.1016/0034-4877(72)90010-9.

[45] Heinz-Peter Breuer, Elsi-Mari Laine, and Jyrki Piilo. Measure for the degree of non-markovian behavior of quantum processes in open systems. *Physical Review Letters*, 103:210401, 2009. doi: 10.1103/PhysRevLett.103.210401.

[46] Ángel Rivas, Susana F. Huelga, and Martin B. Plenio. Quantum non-markovianity: characterization, quantification and detection. *Reports on Progress in Physics*, 77(9):094001, 2014. doi: 10.1088/0034-4885/77/9/094001.

[47] Ryszard Horodecki, Paweł Horodecki, Michał Horodecki, and Karol Horodecki. Quantum entanglement. *Reviews of Modern Physics*, 81(2):865–942, 2009. doi: 10.1103/RevModPhys.81.865.

[48] Luigi Amico, Rosario Fazio, Andreas Osterloh, and Vlatko Vedral. Entanglement in many-body systems. *Reviews of Modern Physics*, 80(2):517–576, 2008. doi: 10.1103/RevModPhys.80.517.

[49] Kenneth S Krane. *Introductory nuclear physics*. John Wiley & Sons, 1991.

[50] Alan Robert Edmonds. *Angular momentum in quantum mechanics*, volume 4. Princeton university press, 1996.

[51] Edward Uhler Condon and George Hiram Shortley. *The theory of atomic spectra*. Cambridge University Press, 1935.